Elementary Analysis

Chapter 2: Sections 13–16

All 47 exercises, with every subpart
Detailed worked solutions

Mathjock

Text and numbering. Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Chapter 2. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

Reading philosophy. Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

Conventions. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, sequence indices run through $\mathbb{N}$ and all terms and finite limits are real. The empty sum is 0, and $0! = 1$.

Coverage. Sections 13–16: Exercises 13.1–13.15, 14.1–14.14, 15.1–15.8, and 16.1–16.10.

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13 Some topological concepts in metric spaces

Throughout this section, k is a positive integer. In $\mathbb{R}^k$, d without a subscript denotes Euclidean distance. In an arbitrary metric space (S, d), write

$$B(x, r) = \{y \in S : d(y, x) < r\} \qquad (r > 0).$$

The ambient space matters: complements, interiors, and closures are taken in the space specified in the exercise. We write $\overline{E}$ for the book's closure E^-, and E° for its interior.

We will use the following elementary ball fact. If $y \in B(x, r)$, then $\rho = r - d(y, x) > 0$. For $z \in B(y, \rho)$, the triangle inequality gives

$$d(z, x) \le d(z, y) + d(y, x) < \rho + d(y, x) = r.$$

Thus $B(y, \rho) \subseteq B(x, r)$, which proves that every open ball in every metric space is open. No assertion about the boundary of a ball in a general metric space is being assumed.

13.1 Exercise 13.1: Two complete metrics on finite-dimensional space

What we must show. For $x, y \in \mathbb{R}^k$, verify that

$$d_1(x, y) = \max_{1 \le j \le k} |x_j - y_j|, \qquad d_2(x, y) = \sum_{j=1}^{k} |x_j - y_j|$$

are metrics, and prove that each metric is complete.

Plan. Check the three metric axioms directly. For completeness, first obtain a limit of each coordinate, then combine the finitely many coordinate estimates using one cutoff.

(a), first axiom: finite nonnegative distances, with the correct zeros.

Each $|x_j - y_j|$ is a finite nonnegative real number. A finite maximum and a finite sum of these quantities are therefore finite and nonnegative. If $x = y$, every coordinate difference is zero, and both distances are zero. Conversely, if the maximum is zero, then every nonnegative coordinate difference is at most zero and so is zero. If the sum is zero, every nonnegative summand must also be zero. In either case $x_j = y_j$ for all j, hence $x = y$. Equivalently, if $x \neq y$, at least one difference is positive, and both distances are positive.

(a), second and third axioms: symmetry and triangle inequalities.

The identity $|x_j - y_j| = |y_j - x_j|$ gives symmetry for both formulas. For any $x, y, z \in \mathbb{R}^k$, the real triangle inequality gives

$$|x_j - z_j| \le |x_j - y_j| + |y_j - z_j|.$$

For d_1, the right side is at most $d_1(x, y) + d_1(y, z)$ for every j. Taking the maximum of the left sides proves $d_1(x, z) \le d_1(x, y) + d_1(y, z)$. For d_2, summing the coordinate inequalities gives

$$d_2(x, z) \le \sum_{j=1}^{k} |x_j - y_j| + \sum_{j=1}^{k} |y_j - z_j| = d_2(x, y) + d_2(y, z).$$

This checks all metric axioms.

(b), coordinate limits: start with a Cauchy sequence in either metric.

Let $(x^{(n)})$ be Cauchy for d_i, where $i = 1$ or $i = 2$. For every coordinate j,

$$\left|x_j^{(m)} - x_j^{(n)}\right| \leq d_i(x^{(m)}, x^{(n)}).$$

For d_1, this is the defining maximum bound; for d_2, it is the bound of one nonnegative summand by the whole sum. Given $\varepsilon > 0$, the Cauchy cutoff for d_i therefore also makes every coordinate difference smaller than ε. Each coordinate sequence is Cauchy in $\mathbb{R}$. Completeness of $\mathbb{R}$ gives a real limit x_j for each j. Put $x = (x_1, \ldots, x_k) \in \mathbb{R}^k$.

(b), convergence in the maximum metric.

Fix $\varepsilon > 0$. For each j, choose N_j so that $n > N_j$ implies $\left|x_j^{(n)} - x_j\right| < \varepsilon$. Take $N = \max\{N_1, \ldots, N_k\}$. If $n > N$, every one of the finitely many differences is strictly less than ε. Their maximum is therefore strictly less than ε: $d_1(x^{(n)}, x) < \varepsilon$.

(b), convergence in the sum metric.

For d_2, instead choose N_j to give coordinate errors smaller than ε/k, which is positive because $k \geq 1$. Beyond the maximum of these cutoffs,

$$d_2(x^{(n)}, x) = \sum_{j=1}^{k} \left|x_j^{(n)} - x_j\right| < k\frac{\varepsilon}{k} = \varepsilon.$$

Thus a Cauchy sequence for either metric has a limit in that same metric and in the same space $\mathbb{R}^k$.

Conclusion. Both d_1 and d_2 are complete metrics on $\mathbb{R}^k$.

13.2 Exercise 13.2: Euclidean distance and convergence of coordinates

What we must show. Prove both inequalities in Lemma 13.3(1), then prove that a sequence in $\mathbb{R}^k$ converges if and only if every coordinate sequence converges in $\mathbb{R}$.

(a) Compare one summand, the sum, and its largest summand.

For $x, y \in \mathbb{R}^k$, let $M = \max_{1 \leq i \leq k} |x_i - y_i|$. Every square is nonnegative, so for each j,

$$|x_j - y_j|^2 \leq \sum_{i=1}^{k} |x_i - y_i|^2 \leq kM^2.$$

The first inequality retains one summand and discards only nonnegative summands. The second replaces each of the k summands by the upper bound M^2. Taking nonnegative square roots, which preserves order, gives

$$\boxed{|x_j - y_j| \leq d(x, y) \leq \sqrt{k} \max_{1 \leq i \leq k} |x_i - y_i|\,.}$$

(b), forward direction: Euclidean convergence gives coordinate convergence.

Suppose $x^{(n)} \to x$ in Euclidean distance. Given $\varepsilon > 0$, there is N such that $d(x^{(n)}, x) < \varepsilon$ for $n > N$. The first inequality in (a) then gives $\left|x_j^{(n)} - x_j\right| < \varepsilon$ for every j and $n > N$. Thus each $x_j^{(n)} \to x_j$ in $\mathbb{R}$.

(b), reverse direction: select one cutoff for all the coordinates.

Suppose $x_j^{(n)} \to x_j \in \mathbb{R}$ for $j = 1, \ldots, k$, and define $x = (x_1, \ldots, x_k)$. Given $\varepsilon > 0$, choose N_j such that

$$n > N_j \quad \Longrightarrow \quad \left|x_j^{(n)} - x_j\right| < \frac{\varepsilon}{\sqrt{k}}.$$

Set $N = \max_j N_j$. For $n > N$, the maximum of all coordinate errors is smaller than $\varepsilon/\sqrt{k}$. The second inequality in (a) gives $d(x^{(n)}, x) < \varepsilon$.

Conclusion. The distance bounds are proved, and coordinate convergence is equivalent to Euclidean convergence.

13.3 Exercise 13.3: The supremum metric on bounded sequences

What we must show. Let B be the set of all bounded real sequences. Prove that

$$d(x, y) = \sup_{j \in \mathbb{N}} |x_j - y_j|$$

is a metric on B, and decide whether the analogous infinite sum defines a metric on all of B.

(a) First verify that the supremum is a finite real number.

For $x, y \in B$, choose finite nonnegative bounds C_x, C_y with $|x_j| \le C_x$ and $|y_j| \le C_y$ for all j. Then

$$0 \le |x_j - y_j| \le |x_j| + |y_j| \le C_x + C_y.$$

The set of coordinate differences is nonempty and bounded above, so its supremum exists and belongs to $[0, C_x + C_y]$.

(a) Check the zeros, symmetry, and triangle inequality.

If $x = y$, every difference is zero and so is the supremum. If $d(x, y) = 0$, then each $0 \le |x_j - y_j| \le 0$, implying $x_j = y_j$ for all j, hence $x = y$. Symmetry follows coordinate by coordinate from symmetry of absolute value. Finally, for every j,

$$|x_j - z_j| \le |x_j - y_j| + |y_j - z_j| \le d(x, y) + d(y, z).$$

The final expression is an upper bound for all coordinate differences between x and z. Their least upper bound is thus at most that expression. This is the triangle inequality for d.

(b) A sum can be infinite even when both sequences are bounded.

Take $x = (0, 0, 0, \ldots)$ and $y = (1, 1, 1, \ldots)$. Both belong to B, but

$$\sum_{j=1}^{\infty} |x_j - y_j| = \sum_{j=1}^{\infty} 1 = +\infty,$$

since its nth partial sum is n. A metric in Definition 13.1 must give a finite real distance between each pair of points.

Conclusion. The supremum formula is a metric on B. The sum formula is not a metric on all of B, because it need not be finite.

13.4 Exercise 13.4: Unions and finite intersections of open sets

What we must show. Prove that arbitrary unions and finite intersections of open subsets of a metric space (S, d) are open.

(iii) For a point of a union, use one set that contains it.

Let $U = \bigcup_{\lambda\in\Lambda} U_\lambda$, where each U_λ is open. If $x \in U$, there is a λ_0 with $x \in U_{\lambda_0}$. Since that set is open, some $r > 0$ satisfies $B(x, r) \subseteq U_{\lambda_0} \subseteq U$. Thus every point of U is interior, which says U is open. If the collection is empty, its union is the empty set. That set is open because there is no point for which the interior condition could fail.

(iv) For a point of a finite intersection, take the smallest radius.

Let $V = U_1 \cap \cdots \cap U_m$, where $m \geq 1$ and all the U_i are open. For $x \in V$, choose $r_i > 0$ with $B(x, r_i) \subseteq U_i$. The number $r = \min\{r_1, \ldots, r_m\}$ is positive because this is a finite nonempty set of positive numbers. If $d(y, x) < r$, then $d(y, x) < r_i$ for every i, so $y \in U_i$ for every i. Hence $B(x, r) \subseteq V$, proving openness. The intersection of no subsets of S is S, which is open: every ball with center in S is a subset of S.

An infinite intersection need not be open: in $\mathbb{R}$, $\bigcap_{n\geq 1}(-1/n, 1/n) = \{0\}$, and no positive-radius interval about 0 fits in $\{0\}$. This explains why the intersection statement requires finiteness.

Conclusion. Arbitrary unions and finite intersections preserve openness.

13.5 Exercise 13.5: De Morgan's law and intersections of closed sets

What we must show. Verify the stated complement identity and deduce that an arbitrary intersection of closed subsets of S is closed.

(a) Prove equality by testing membership of a point.

For any $x \in S$, the following statements are equivalent:

$$\begin{aligned} x \in \bigcap_{U\in\mathcal{U}} (S \setminus U) \quad &\iff \quad x \notin U \text{ for every } U \in \mathcal{U} \\ &\iff \quad x \notin \bigcup_{U\in\mathcal{U}} U \\ &\iff \quad x \in S \setminus \bigcup_{U\in\mathcal{U}} U. \end{aligned}$$

Since every point has the same membership on the two sides, the sets are equal. For the empty collection, the left side is S and the right side is $S \setminus \varnothing = S$, so the convention is consistent.

(b) Take complements, then apply the open-union result.

Let F_λ be closed subsets of S, and put $F = \bigcap_{\lambda\in\Lambda} F_\lambda$. A point fails to belong to this intersection exactly when it fails to belong to at least one F_λ. Thus

$$S \setminus F = \bigcup_{\lambda\in\Lambda} (S \setminus F_\lambda).$$

Each complement on the right is open by the definition of a closed set. Their union is open by Exercise 13.4, so F is closed by that same definition. If Λ is empty, $F = S$ and its complement is empty and open.

Conclusion. The complement identity holds, and every intersection of closed sets is closed.

13.6 Exercise 13.6: Four equivalent descriptions involving closure

What we must show. Prove all four assertions of Proposition 13.9 for a subset E of an arbitrary metric space (S, d).

Plan. First derive a ball characterization directly from the definition of closure as an intersection of closed supersets. Then obtain the sequence characterizations without assuming them.

Preparation: closure is a closed superset, and a ball detects its points.

The family of closed sets containing E is nonempty, because S belongs to it. Every set in that family contains E; therefore their intersection $\overline{E}$ contains E. Exercise 13.5 shows that the intersection is closed.

For any $x \in S$, we claim

$$x \in \overline{E} \quad \Longleftrightarrow \quad B(x, r) \cap E \neq \varnothing \text{ for every } r > 0. \tag{*}$$

If some ball $B(x, r)$ misses E, then its complement is a closed set containing E, since balls are open. This closed superset excludes x, so $x \notin \overline{E}$. Conversely, if $x \notin \overline{E}$, then at least one closed set C containing E excludes x; otherwise x would belong to their intersection. The open set $S \setminus C$ contains x, so it contains some $B(x, r)$, and that ball misses E. These two implications prove (*).

(a) A set is closed exactly when it equals its closure.

If E is closed, it is one of the closed supersets in the defining intersection, so $\overline{E} \subseteq E$. The reverse inclusion was already proved, giving equality. If $E = \overline{E}$, then E is closed because $\overline{E}$ was proved closed. This proves both directions.

(c) Closure points are exactly limits of sequences of points of E.

We establish (c) before (b) so that there is no circular argument. If $x \in \overline{E}$, apply (*) with radius $1/n$ to choose $x_n \in E$ with $d(x_n, x) < 1/n$. For any $\varepsilon > 0$, choose an integer $N > 1/\varepsilon$. If $n > N$, then $d(x_n, x) < 1/n < \varepsilon$. Hence $x_n \to x$.

Conversely, if $x_n \in E$ and $x_n \to x \in S$, fix $r > 0$. Convergence gives some index n with $d(x_n, x) < r$, so $x_n \in B(x, r) \cap E$. Every ball about x thus meets E, and (*) gives $x \in \overline{E}$. For $E = \varnothing$, no such sequence exists and its closure is empty, so the equivalence still holds.

(b) Closed sets are exactly those containing limits of their convergent sequences.

If E is closed and $x_n \in E$ tends to a point $x \in S$, part (c) gives $x \in \overline{E}$, and part (a) gives $\overline{E} = E$, so $x \in E$. Conversely, suppose E contains every such limit. For every $x \in \overline{E}$, part (c) supplies a sequence in E converging to x, so the assumption gives $x \in E$. Thus $\overline{E} \subseteq E$. Combined with the always-valid inclusion $E \subseteq \overline{E}$, this gives equality, and part (a) makes E closed.

(d) Boundary points see both the set and its complement.

By definition, the boundary is $\overline{E} \setminus E^{\circ}$. A point x lies outside E° exactly when no ball about x is contained in E. Indeed, the existence of such a ball would also put x in E, since its center belongs to the ball, and would make x interior. Saying no ball is contained in E is equivalent to saying every

ball meets $S \setminus E$. By (*) applied to this complement,

$$x \notin E^\circ \quad \Longleftrightarrow \quad x \in \overline{S \setminus E}.$$

Consequently

$$\overline{E} \setminus E^\circ = \overline{E} \cap \overline{S \setminus E},$$

which is precisely the assertion in (d).

Conclusion. All four claims follow from the closure definition and the ball criterion (*), with no sequence characterization assumed in advance.

13.7 Exercise 13.7: Decomposing an open real set into disjoint intervals

What we must show. Express every open $U \subseteq \mathbb{R}$ as a finite or countably infinite disjoint union of open intervals. Unbounded intervals are allowed; the empty set is the union of an empty family.

1. Group points that have no gap of U between them.

For $x, y \in U$, write $x \sim y$ when the whole closed segment with endpoints x, y lies in U. This relation is reflexive, because $[x, x] = \{x\} \subseteq U$, and symmetric because the segment is unchanged when its endpoints are interchanged. For transitivity, if $x \sim y$ and $y \sim z$, every point between x and z lies in the union of the segments joining x to y and y to z. That union is contained in U. Thus $x \sim z$, so $\sim$ is an equivalence relation. Its classes partition U into nonempty pairwise disjoint sets.

2. Show that each class is an interval and is open.

Let I be a class, and let $p < q$ belong to it. The equivalence relation gives $[p, q] \subseteq U$. For $p < t < q$, the segment $[p, t]$ is contained in $[p, q]$, so $p \sim t$ and $t \in I$. Thus every point between two points of I belongs to I.

For $z \in I$, openness of U gives $r > 0$ with $(z - r, z + r) \subseteq U$. Every t in that interval has its segment to z inside the interval, hence inside U. Therefore $t \sim z$ and $t \in I$. We have proved $(z - r, z + r) \subseteq I$, so I is itself open.

3. Identify the endpoints, including infinite endpoints.

Let $a = \inf I$ and $b = \sup I$ in the extended real sense; use $a = -\infty$ or $b = +\infty$ if the corresponding bound does not exist in $\mathbb{R}$. Since I is nonempty and open, it contains an interval around one of its points, and $a < b$. For $a < t < b$, the definitions of infimum and supremum give $p, q \in I$ with $p < t < q$; otherwise t would be a lower or upper bound contradicting $a < t < b$. The interval property puts $t \in I$. Conversely, a finite endpoint cannot lie in I: openness there would supply a point of I still smaller than a or still larger than b. Hence $I = (a, b)$, with the usual interpretations $(a, +\infty)$, $(-\infty, b)$, or $\mathbb{R}$.

4. Prove that there are at most countably many classes.

Fix an enumeration $q_1, q_2, \ldots$ of $\mathbb{Q}$. Each nonempty open interval I contains a rational by density. Assign to I the least index $m(I)$ for which $q_{m(I)} \in I$. This least index exists by the least-element property of $\mathbb{N}$. Distinct classes have distinct assigned indices, because they are disjoint and cannot contain the same rational. Thus the classes inject into $\mathbb{N}$. If there are infinitely many, list their assigned indices in increasing order and use that order to list the classes; every assigned index appears, since only finitely many positive integers precede it.

Conclusion. The equivalence classes give the required disjoint interval decomposition. The family may be empty, finite, or countably infinite, and intervals may be unbounded.

13.8 Exercise 13.8: Intervals, balls, half-spaces, and examples that are neither

What we must show. Verify the assertions in the first paragraph of Example 3 and all the assertions in Example 4. In the interval assertions assume $a < b$; in the ball assertions assume $r > 0$ and $k \geq 1$. We address the degenerate cases separately below.

(a) Open and closed intervals in the real line.

For $x \in (a, b)$, let $\rho = \min\{x - a, b - x\} > 0$. If $|y - x| < \rho$, then $a < y < b$. Thus (a, b) is open. The complement of $[a, b]$ is $(-\infty, a) \cup (b, +\infty)$. Each ray is open: for a point strictly on its allowed side, half its distance to the endpoint provides a ball staying on that side. Their union is open, so $[a, b]$ is closed.

Every $x \in (a, b)$ is interior to $[a, b]$ by the first estimate. The endpoint a is not interior: every ball about it contains $a - \rho/2 < a$ for its radius $\rho > 0$. Similarly b is not interior because $b + \rho/2$ is outside $[a, b]$. Hence $[a, b]^\circ = (a, b)$.

The closed set $[a, b]$ contains (a, b), so $\overline{(a, b)} \subseteq [a, b]$. Every ball about a meets (a, b): choose a displacement $0 < t < \min\{\rho, b - a\}$ and use $a + t$. The point $b - t$ does the same for b. The ball criterion of Exercise 13.6 therefore puts both endpoints in the closure. Interior points already belong to the set. Thus $\overline{(a, b)} = [a, b] = \overline{[a, b]}$. Subtracting the interior gives boundary $\{a, b\}$ for both intervals.

If $a = b$, the open interval is empty and has empty closure and boundary. The closed interval is the singleton $\{a\}$, which is closed, has empty interior, and has boundary $\{a\}$. This is why the boundary assertion for both intervals presumes $a < b$.

(b), balls: openness, closedness, and the possible boundary points.

For a center $c \in \mathbb{R}^k$, put

$$D = \{x : d(x, c) < r\}, \qquad K = \{x : d(x, c) \leq r\}.$$

The opening ball argument of this section shows D is open. If $x \notin K$, let $\rho = d(x, c) - r > 0$. For $d(y, x) < \rho$, the triangle inequality rearranges to

$$d(y, c) \geq d(x, c) - d(x, y) > d(x, c) - \rho = r.$$

Thus a ball about x stays outside K, so the complement of K is open and K is closed.

If $d(x, c) < r$, then x is interior to both sets, using the radius $r - d(x, c)$. If $d(x, c) > r$, the preceding complement ball shows x lies outside the closure of either set. Only points with $d(x, c) = r$ can therefore be boundary points.

(b), spheres: every point on the sphere really is a boundary point.

Let $d(x, c) = r > 0$, and let $\varepsilon > 0$. Choose $0 < t < \min\{1, \varepsilon/r\}$. Define

$$x_- = c + (1 - t)(x - c), \qquad x_+ = c + (1 + t)(x - c).$$

The Euclidean distance formula gives

$$d(x_-, c) = (1-t)r < r, \quad d(x_+, c) = (1+t)r > r, \quad d(x_-, x) = d(x_+, x) = tr < \varepsilon.$$

Thus every ball about x meets D (at x_-) and the complement of K (at x_+). It meets both either set and its complement, so Exercise 13.6(d) puts x on both boundaries. Consequently both boundaries are exactly $\{x : d(x, c) = r\}$. This radial argument uses Euclidean space; the same sphere assertion is not valid for all metrics. If $r = 0$, the open ball is empty with empty boundary, while the closed ball is the singleton $\{c\}$ with boundary $\{c\}$.

(b), strict coordinate inequalities give open half-spaces and quadrants.

In $\mathbb{R}^2$, consider $H = \{x : x_1 > 0\}$ and $Q = \{x : x_1 > 0,\ x_2 > 0\}$. For $x \in H$, choose $\rho = x_1/2 > 0$. Since $|y_1 - x_1| \le d(y, x)$, every y in $B(x, \rho)$ satisfies $y_1 > x_1 - \rho = x_1/2 > 0$. Thus H is open. For $x \in Q$, choose $\rho = \min\{x_1, x_2\}/2$. The same coordinate estimate, separately for $j = 1, 2$, gives $y_j > x_j - \rho \ge x_j/2 > 0$. Hence Q is open.

For $H_0 = \{x : x_1 \ge 0\}$, its complement is $\{x : x_1 < 0\}$, an open half-space by the same radius argument with $-x_1$ in place of x_1. Thus H_0 is closed. For $Q_0 = \{x : x_1 \ge 0,\ x_2 \ge 0\}$, the complement is $\{x : x_1 < 0\} \cup \{x : x_2 < 0\}$, a union of open half-spaces. Thus Q_0 is closed as asserted.

(b), the half-open interval is neither open nor closed.

The point $0 \in [0, 1)$ is not interior, since every ball about 0 contains negative points. Hence $[0, 1)$ is not open. The sequence $1 - 1/n$ belongs to $[0, 1)$ and tends to 1, which is outside the set. The closed-set sequence criterion of Exercise 13.6(b) shows it is not closed.

(b), the mixed quadrant is neither open nor closed.

Let $M = \{x : x_1 > 0,\ x_2 \ge 0\}$. The point $(1, 0)$ belongs to M, but every ball of radius $\rho > 0$ about it contains $(1, -\rho/2)$, outside M. Thus M is not open. The sequence $(1/n, 0)$ lies in M and tends in Euclidean distance to $(0, 0)$, outside M. It follows that M is not closed either.

Conclusion. All the interval, Euclidean ball, half-space, quadrant, and mixed examples have the stated properties, with the positive radius and nondegenerate interval conditions made explicit.

13.9 Exercise 13.9: Three closures in the real line

What we must show. Find the closures of $\{1/n : n \in \mathbb{N}\}$, $\mathbb{Q}$, and $\{q \in \mathbb{Q} : q^2 < 2\}$ in $\mathbb{R}$.

(a) Include the reciprocal terms and their one additional limit.

Let $A = \{1/n : n \in \mathbb{N}\}$. Every point of A belongs to its closure, and $1/n \to 0$ gives $0 \in \overline{A}$ by Exercise 13.6(c). We show no other point belongs. If $x < 0$, the ball of radius $-x/2$ about x is negative and misses A. If $x > 1$, the radius $(x-1)/2$ gives a ball wholly above 1, which also misses A.

It remains to exclude $0 < x \le 1$ with $x \notin A$. Choose N so large that $1/N < x/2$. All terms $1/n$ for $n \ge N$ are then below $x/2$. The finite set $\{1, 1/2, \ldots, 1/(N-1)\}$ has no member equal to x. Its finitely many positive distances from x have a positive minimum η. Here $N \ge 2$ because $1/N < x/2 \le 1/2$. A ball about x of radius $\min\{x/2, \eta\}/2$ misses the finite set and all the remaining reciprocals. The ball criterion excludes x. Therefore

$$\boxed{\overline{A} = A \cup \{0\}.}$$

(b) Density fills the whole real line.

For any $x \in \mathbb{R}$ and $r > 0$, the interval $(x - r, x + r)$ contains a rational by density of $\mathbb{Q}$. Hence every real x satisfies the closure ball criterion, giving $\boxed{\overline{\mathbb{Q}} = \mathbb{R}}$.

(c) The rational points between the two square-root endpoints.

For real q, the inequality $q^2 < 2$ is equivalent to $-\sqrt{2} < q < \sqrt{2}$. Thus the given set is $C = \mathbb{Q} \cap (-\sqrt{2}, \sqrt{2})$. If $-\sqrt{2} < x < \sqrt{2}$, then every neighborhood of x contains a smaller nonempty interval lying inside $(-\sqrt{2}, \sqrt{2})$; density supplies a rational in that interval. For $x = \sqrt{2}$, an interval $(\sqrt{2} - t, \sqrt{2})$ with $0 < t < \min\{r, 2\sqrt{2}\}$ lies inside the radius-r neighborhood and inside the allowed range, and contains a rational. The interval $(-\sqrt{2}, -\sqrt{2} + t)$ proves the same statement for $x = -\sqrt{2}$. Therefore the whole closed interval belongs to $\overline{C}$.

If $x > \sqrt{2}$, take radius $(x - \sqrt{2})/2$; if $x < -\sqrt{2}$, take radius $(-\sqrt{2} - x)/2$. The resulting ball misses the closed interval and hence misses C. No outside point belongs to the closure, so

$$\boxed{\overline{C} = [-\sqrt{2}, \sqrt{2}].}$$

Conclusion. The closures are respectively $\{0\} \cup \{1/n : n \in \mathbb{N}\}$, $\mathbb{R}$, and $[-\sqrt{2}, \sqrt{2}]$.

13.10 Exercise 13.10: Three sets with no interior points

What we must show. Prove that the reciprocal set, the rationals, and the Cantor set all have empty interior in $\mathbb{R}$.

(a) A rational-only set cannot contain a real interval.

Every reciprocal $1/n$ is rational. Every open interval contains an irrational number. For completeness, to find one in (a, b), choose a rational q in $(a - \sqrt{2}, b - \sqrt{2})$ by rational density; then $q + \sqrt{2} \in (a, b)$ is irrational, since its being rational would make $\sqrt{2} = (q + \sqrt{2}) - q$ rational. Thus every ball about any point of the reciprocal set contains a point outside that set. No point of the set is interior.

(b) Apply the same irrational-density argument to $\mathbb{Q}$.

For any $q \in \mathbb{Q}$ and $r > 0$, the preceding argument gives an irrational in $(q - r, q + r)$. That ball is not a subset of $\mathbb{Q}$. Thus $\mathbb{Q}$ also has no interior points. Notice that dense closure does not imply nonempty interior: Exercise 13.9 gave $\overline{\mathbb{Q}} = \mathbb{R}$.

(c) The intervals remaining at late Cantor stages are too short.

Use the numbering in Example 5: $F_1 = [0, 1]$, and F_n is the union of 2^{n-1} disjoint closed intervals, each of length $3^{-(n-1)}$. The Cantor set is $F = \bigcap_{n \geq 1} F_n$. Suppose, for contradiction, that $x \in F^\circ$. Then some real interval $(x - r, x + r)$, with $r > 0$, lies in F. The closed interval $J = [x - r/2, x + r/2]$ has length r and also lies in F, hence in every F_n.

Choose n so large that $3^{-(n-1)} < r$. The interval J would have to lie in one of the component intervals of F_n. Indeed, if it met two distinct components, it would also contain the points of the nonempty removed gap between those components, because it contains every point between any two of its points. That would contradict $J \subseteq F_n$. But an interval of length r cannot be contained in an interval of smaller length $3^{-(n-1)}$. This is a contradiction.

Conclusion. All three interiors are empty. The Cantor proof uses the explicit shrinking intervals of its construction.

13.11 Exercise 13.11: Compactness and subsequences in Euclidean space

What we must show. Show that $E \subseteq \mathbb{R}^k$ is compact if and only if every sequence of points of E has a subsequence converging to a point of E.

Plan. Use Heine–Borel and Bolzano–Weierstrass. For the converse, show that the stated subsequence property forces both boundedness and closedness.

Preparation: uniqueness and inheritance of metric limits.

If $x_n \to x$ and $x_n \to y$, then $d(x,y) \le d(x,x_n) + d(x_n,y) \to 0$, so $d(x,y) = 0$ and $x = y$. Every subsequence of a convergent sequence has the same limit: strictly increasing selection indices n_j satisfy $n_j \ge j$, so every eventual distance estimate for the original sequence also holds for all sufficiently late selected terms. Also, a convergent sequence in $\mathbb{R}^k$ is bounded: its tail lies within distance 1 of its limit x, and for each coordinate the triangle inequality gives $\left|x_j^{(n)}\right| \le |x_j| + 1$ on that tail. A finite maximum accommodates the omitted initial coordinates.

Forward direction: combine boundedness, subsequences, and closedness.

If E is compact, Heine–Borel (Theorem 13.12) says it is closed and bounded. Any sequence in E is then bounded. Bolzano–Weierstrass (Theorem 13.5) gives a subsequence tending to some point of $\mathbb{R}^k$. Since all its terms lie in the closed set E, Exercise 13.6(b) puts that limit in E.

Reverse direction, boundedness: unbounded points would escape every subsequence.

Assume the stated subsequence property. If E were unbounded, then for each $n \in \mathbb{N}$ we could choose $x^{(n)} \in E$ with

$$\max_{1 \le j \le k} \left|x_j^{(n)}\right| > n.$$

For any subsequence $x^{(n_m)}$, the selected indices satisfy $n_m \ge m$, so its coordinate maximum exceeds m. It is therefore unbounded: for any proposed finite bound C, take $m > C$. It cannot converge, since a convergent sequence was just shown to be bounded. This contradicts the assumed property. Thus E is bounded.

Reverse direction, closedness: recover every closure point.

Let $x \in \overline{E}$. By Exercise 13.6(c), choose a sequence $x_n \in E$ converging to x. The assumed property supplies a subsequence $x_{n_m} \to y$ for some $y \in E$. Inheritance gives $x_{n_m} \to x$ as well, and uniqueness of the limit gives $x = y \in E$. Thus $\overline{E} \subseteq E$. The reverse inclusion always holds, so Exercise 13.6(a) makes E closed. Having proved both closedness and boundedness, Heine–Borel gives compactness.

If $E = \varnothing$, it is compact because the empty subfamily is a finite subcover of every cover of it. There is no sequence of points of E, so the sequential condition is vacuously true.

Conclusion. The two conditions are equivalent for all subsets of $\mathbb{R}^k$, including the empty set.

13.12 Exercise 13.12: Closed subsets and finite unions of compact sets

What we must show. In any metric space (S, d), prove that a closed subset of a compact set is compact, and that a finite union of compact sets is compact.

Plan. Work directly with open covers, so neither result depends on the finite-dimensional Heine–Borel theorem.

(a) Add one open set to cover the rest of the larger compact set.

Suppose $E \subseteq F$, E is closed in S, and F is compact. Let $\mathcal{U}$ be any family of open subsets of S covering E. The set $V = S \setminus E$ is open, and $\mathcal{U} \cup \{V\}$ covers F: a point of F either belongs to E and is covered by $\mathcal{U}$, or lies in V. Compactness of F gives a finite subfamily covering F. Remove V if it was selected. The remaining sets still cover E, since no point of E belongs to V. They are a finite subfamily of $\mathcal{U}$, so E is compact.

If "closed subset of F" is interpreted as closed relative to F, the same conclusion holds: relative closedness means $F \setminus E = F \cap V$ for some open $V \subseteq S$. Use that V in the preceding cover argument; it still misses E and covers every point of $F \setminus E$.

(b) Take a finite subcover for each compact piece.

Let $K = K_1 \cup \cdots \cup K_m$ with each K_i compact, and let $\mathcal{U}$ be an open cover of K. It also covers each K_i. For each i, select a finite subfamily $\mathcal{U}_i \subseteq \mathcal{U}$ covering K_i. The family $\mathcal{U}_1 \cup \cdots \cup \mathcal{U}_m$ is finite, since a finite union of finite sets is finite, and it covers every piece, hence their union K. This proves compactness. When $m = 0$, the union is empty and is compact by the empty-subcover argument.

Conclusion. Both permanence properties hold in every metric space.

13.13 Exercise 13.13: A compact real set contains its extreme bounds

What we must show. For a nonempty compact $E \subseteq \mathbb{R}$, show $\sup E \in E$ and $\inf E \in E$.

1. Ensure the bounds are finite and use the approximation property.

Heine–Borel makes E closed and bounded. Since it is also nonempty, completeness of $\mathbb{R}$ supplies finite real numbers $M = \sup E$ and $m = \inf E$. For every $n \in \mathbb{N}$, $M - 1/n$ is strictly below the least upper bound, so it cannot be an upper bound. Choose $u_n \in E$ with

$$M - \frac{1}{n} < u_n \leq M.$$

Similarly, $m + 1/n$ is strictly above the greatest lower bound, so it cannot be a lower bound. Choose $v_n \in E$ with

$$m \leq v_n < m + \frac{1}{n}.$$

2. Check convergence, then use closedness.

The first inequalities give $|u_n - M| < 1/n$, so $u_n \to M$. The second give $|v_n - m| < 1/n$, so $v_n \to m$. Both sequences lie in E. Closedness and Exercise 13.6(b) therefore put both limits in E.

Conclusion. The supremum and infimum are attained: E has a maximum and a minimum, namely M and m.

13.14 Exercise 13.14: The diameter of a compact set is attained

What we must show. For nonempty compact $E \subseteq \mathbb{R}^k$, put

$$\delta = \sup\{d(x, y) : x, y \in E\}.$$

Find $x^0, y^0 \in E$ such that $d(x^0, y^0) = \delta$.

Plan. Approximate the supremum by pairs, select subsequences for both entries of the pairs, and justify passage to the limit using the triangle inequality.

1. Verify that the distance supremum is a finite real number.

Heine–Borel makes E bounded. Choose $C \geq 0$ so that every coordinate of every $x \in E$ has absolute value at most C. For $x, y \in E$, each $|x_j - y_j| \leq 2C$, so $d(x, y) \leq 2C\sqrt{k}$ by Exercise 13.2(a). The set of distances is nonempty because E is nonempty, and contains $d(x, x) = 0$. Its supremum therefore exists with $0 \leq \delta \leq 2C\sqrt{k}$.

2. Choose pairs whose distances approach the supremum.

For each $n \in \mathbb{N}$, the real number $\delta - 1/n$ is not an upper bound of the distance set. Hence choose $x_n, y_n \in E$ such that

$$\delta - \frac{1}{n} < d(x_n, y_n) \leq \delta.$$

This works even when $\delta = 0$; any pair then has distance $0 > \delta - 1/n$. The displayed error bound gives $d(x_n, y_n) \to \delta$.

3. Pass to a common subsequence for the two endpoints.

By Exercise 13.11, some subsequence x_{n_j} tends to a point $x^0 \in E$. Apply the same result to the sequence y_{n_j} in E to find further indices j_ℓ with $y_{n_{j_\ell}} \to y^0 \in E$. The matching $x_{n_{j_\ell}}$ still tends to x^0, since a subsequence inherits a limit. This further selection is also a subsequence of the original sequence of pairs because the composition of two increasing index selections is increasing.

4. Prove that distances pass to the limit.

Two uses of the triangle inequality give

$$d(x, y) \leq d(x, x^0) + d(x^0, y^0) + d(y^0, y),$$
$$d(x^0, y^0) \leq d(x^0, x) + d(x, y) + d(y, y^0).$$

Subtracting $d(x^0, y^0)$ in the first and $d(x, y)$ in the second combines into

$$\Big|d(x, y) - d(x^0, y^0)\Big| \leq d(x, x^0) + d(y, y^0).$$

At the selected pairs, the right side tends to zero. Thus $d(x_{n_{j_\ell}}, y_{n_{j_\ell}}) \to d(x^0, y^0)$. The same distances tend to δ, being a subsequence of the distance sequence in Step 2. Uniqueness of real limits gives $d(x^0, y^0) = \delta$.

Conclusion. Two points of E realize its diameter, including the zero-diameter case.

13.15 Exercise 13.15: A closed bounded set that is not compact

What we must show. In the bounded-sequence space (B, d) of Exercise 13.3, let

$$F = \left\{ x \in B : \sup_{j \in \mathbb{N}} |x_j| \leq 1 \right\}.$$

Show that F is closed and bounded, but not compact.

Plan. Verify closedness using an open complement. Then use the prescribed cover by open balls of radius 1, together with infinitely many points whose pairwise distance is 2.

(a), boundedness: use the zero sequence as the center.

The sequence $0 = (0, 0, \ldots)$ belongs to B. For $x \in F$,

$$d(x, 0) = \sup_j |x_j| \leq 1.$$

Thus F lies in the closed metric ball with center 0 and radius 1, which is exactly the boundedness condition stated in the exercise.

(a), closedness: exhibit a ball about each point outside F.

If $x \in B \setminus F$, then $\sup_j |x_j| > 1$. At least one coordinate, say j_0, satisfies $|x_{j_0}| > 1$; otherwise 1 would be an upper bound for all these absolute values and hence for their supremum. Set $\rho = (|x_{j_0}| - 1)/2 > 0$. If $d(x, y) < \rho$, the real triangle inequality gives

$$|y_{j_0}| \geq |x_{j_0}| - |x_{j_0} - y_{j_0}| > |x_{j_0}| - \rho = \frac{|x_{j_0}| + 1}{2} > 1.$$

Such a y lies outside F. Hence a ball about each point of the complement stays in the complement, proving it open and F closed in B.

(b), the open cover: give every point its radius-one ball.

For $x \in F$, define $U(x) = \{y \in B : d(y, x) < 1\}$. These sets are open in B by the general ball fact at the start of this section. They cover F, since $d(x, x) = 0 < 1$ puts each $x \in F$ in its own $U(x)$.

(b), separated points: compute the distance exactly.

For each $n \in \mathbb{N}$, let $x^{(n)}$ have coordinate -1 at index n and coordinate 1 at every other index. Every one of its coordinates has absolute value 1, so $x^{(n)} \in F$. If $m \neq n$, the two sequences differ by 2 in absolute value at coordinates m and n, and agree at all other coordinates. Therefore

$$d(x^{(m)}, x^{(n)}) = 2.$$

No ball $U(x)$ can contain two distinct such points: if it contained both, then the metric triangle inequality would give

$$2 = d(x^{(m)}, x^{(n)}) \leq d(x^{(m)}, x) + d(x, x^{(n)}) < 1 + 1 = 2,$$

an impossibility. The strict inequalities in the open-ball definition are essential here.

(b), exclude every finite subcover.

A union of q members of this cover can contain at most q of the separated points, since each member contains at most one. But F contains the infinitely many distinct points $x^{(1)}, x^{(2)}, \ldots$. For example, among its first $q+1$ separated points at least one must be missed. Thus no finite subfamily covers F.

Conclusion. F is closed and bounded in the specified metric space, but this explicit open cover has no finite subcover, so F is not compact. Heine–Borel's closed-and-bounded criterion is specific to Euclidean space and does not apply to all metric spaces.

14 Series

For a sequence (a_n), its Nth partial sum is $S_N = \sum_{n=1}^{N} a_n$. The series $\sum_{n=1}^{\infty} a_n$ converges precisely when (S_N) has a finite real limit. A limit of $+\infty$ or $-\infty$ is divergence, even though it specifies a direction of divergence.

The necessary condition $a_n \to 0$ is not sufficient for convergence. The Comparison, Ratio, and Root Tests supply sufficient conditions in the situations specified below. When terms are nonnegative, convergence and absolute convergence are the same assertion. Unless a lower index is explicitly given, the series in the catalogs begin at $n = 1$; changing finitely many initial terms does not affect convergence, as proved in Exercise 14.9. The logarithm is the natural logarithm.

14.1 Exercise 14.1: Six initial convergence tests

What we must show. Decide convergence for each of the six given series, with a reason that checks the hypotheses of the chosen test.

(a) The series with term $n^4/2^n$.

Let $a_n = n^4/2^n > 0$. Since no term is zero, the successive ratio is well defined. Canceling powers gives

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)^4}{2^{n+1}} \frac{2^n}{n^4} = \frac{1}{2}\left(1 + \frac{1}{n}\right)^4 \longrightarrow \frac{1}{2}.$$

The last limit follows from $1/n \to 0$ and the finite product limit law. Its value is strictly below 1, so the Ratio Test proves absolute convergence. Since the terms are positive, this proves convergence of the requested series.

(b) The series with term $2^n/n!$.

Put $a_n = 2^n/n! > 0$. The identity $(n+1)! = (n+1)n!$ yields

$$\frac{a_{n+1}}{a_n} = \frac{2^{n+1}}{(n+1)!} \frac{n!}{2^n} = \frac{2}{n+1} \longrightarrow 0.$$

Here $0 < 1$, so the Ratio Test proves absolute convergence.

(c) The series with term $n^2/3^n$.

With $a_n = n^2/3^n > 0$, the same cancellation gives

$$\frac{a_{n+1}}{a_n} = \frac{1}{3}\left(1 + \frac{1}{n}\right)^2 \longrightarrow \frac{1}{3} < 1.$$

The Ratio Test again gives absolute convergence. The power of n in the numerator does not prevent this conclusion because the computed ratio limit is strictly less than 1.

(d) The series with term $n!/(n^4+3)$.

Let $a_n = n!/(n^4+3) > 0$. Its ratio is

$$\frac{a_{n+1}}{a_n} = (n+1)\frac{n^4+3}{(n+1)^4+3} = (n+1)\frac{1+3/n^4}{(1+1/n)^4+3/n^4}.$$

The second factor tends to 1, since its numerator and denominator both tend to 1 and the denominator is never zero. In particular, it exceeds $1/2$ eventually. Hence the displayed ratio is eventually at least $(n+1)/2$, and therefore tends to $+\infty$. The divergent part of the Ratio Test applies, since the ratio limit inferior is greater than 1.

We can also see why the term condition fails. There is N after which every ratio is at least 2. Induction then gives $a_{N+k} \geq 2^k a_N$, with $a_N > 0$, so the terms tend to $+\infty$ rather than to zero. The positive partial sums are unbounded, and the series diverges to $+\infty$.

(e) The series with term $\cos^2 n/n^2$.

The trigonometric bound $-1 \leq \cos n \leq 1$ gives $0 \leq \cos^2 n \leq 1$. Since $n^2 > 0$, division preserves order:

$$0 \leq \frac{\cos^2 n}{n^2} \leq \frac{1}{n^2}.$$

The comparison series $\sum 1/n^2$ converges by the p-series criterion with $p = 2 > 1$. The Comparison Test therefore proves convergence of the given nonnegative series.

(f) The series with term $1/\log n$, beginning at $n = 2$.

For $n \geq 2$, the logarithm is positive. The elementary logarithm bound is

$$0 < \log n = \int_1^n \frac{dt}{t} \leq \int_1^n 1\,dt = n - 1 < n.$$

The integral comparison here only justifies the inequality for one logarithm; no integral test for a series is being invoked. Taking reciprocals of positive quantities reverses the inequality, so $1/\log n \geq 1/n$. The harmonic series diverges to $+\infty$, and deleting its first term leaves divergent partial sums. Thus the Comparison Test proves divergence to $+\infty$.

Conclusion. Parts (a), (b), (c), and (e) converge absolutely. Parts (d) and (f) diverge to $+\infty$.

14.2 Exercise 14.2: Seven further convergence tests

What we must show. Classify the seven series, explaining the actual numerator and denominator in each term before applying a test.

(a) The series with term $(n-1)/n^2$.

All terms are nonnegative. For $n \geq 2$, the inequality $n - 1 \geq n/2$ follows by subtracting $n/2$ from both sides: $n/2 - 1 \geq 0$. Therefore

$$\frac{n-1}{n^2} \geq \frac{n/2}{n^2} = \frac{1}{2n} \qquad (n \geq 2).$$

The series $\sum_{n=2}^{\infty} 1/(2n)$ is one half of a tail of the harmonic series, so it diverges to $+\infty$. The Comparison Test proves divergence to $+\infty$ of the requested series. The first term is zero and has no effect on this lower bound for the later partial sums.

(b) The series with term $(-1)^n$.

Every term has absolute value 1, so the terms do not tend to zero. Corollary 14.5 therefore rules out convergence. Directly, the partial sums are $-1, 0, -1, 0, \ldots$: an odd number of terms gives -1, and

each pair $(-1)+1$ sums to zero. Thus the partial sums have two different subsequential limits and do not tend to either infinity.

(c) The series with term $3n/n^3$.

Since $n \neq 0$, cancellation gives $3n/n^3 = 3/n^2$. The p-series $\sum 1/n^2$ converges, and multiplication of all partial sums by the constant 3 preserves convergence by the constant multiple limit law. Hence this positive series converges absolutely.

(d) The series with term $n^3/3^n$.

For $a_n = n^3/3^n > 0$, compute

$$\frac{a_{n+1}}{a_n} = \frac{1}{3}\left(1+\frac{1}{n}\right)^3 \longrightarrow \frac{1}{3} < 1.$$

All ratios are defined; the Ratio Test gives absolute convergence.

(e) The series with term $n^2/n!$.

Put $a_n = n^2/n! > 0$. Cancel $n!$ and one factor of $n+1$:

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)^2}{(n+1)!}\frac{n!}{n^2} = \frac{n+1}{n^2} = \frac{1}{n} + \frac{1}{n^2} \longrightarrow 0 < 1.$$

The Ratio Test proves absolute convergence.

(f) The series with term $1/n^n$.

The terms are positive and their positive nth roots satisfy

$$\left(\frac{1}{n^n}\right)^{1/n} = \frac{1}{n} \longrightarrow 0.$$

Thus the limit superior of the roots is $0 < 1$. The convergent part of the Root Test proves absolute convergence.

(g) The series with term $n^n/2^n$.

Here the positive nth root is

$$\left(\frac{n^n}{2^n}\right)^{1/n} = \frac{n}{2} \longrightarrow +\infty.$$

The root limit superior is greater than 1, so the Root Test proves divergence. More explicitly, for $n \geq 4$ one has $n/2 \geq 2$, so $(n/2)^n \geq 2^n$. These positive terms tend to $+\infty$; hence the partial sums do too.

Conclusion. Parts (c), (d), (e), and (f) converge absolutely. Parts (a) and (g) diverge to $+\infty$. Part (b) diverges with bounded, oscillating partial sums.

14.3 Exercise 14.3: Factorials, geometric comparisons, and a periodic term

What we must show. Classify each of the six series and justify convergence or failure of the necessary term condition.

(a) The series with term $1/\sqrt{n!}$.

Set $a_n = 1/\sqrt{n!} > 0$. The factorial is inside the square root. Using $(n+1)! = (n+1)n!$ and positivity,

$$\frac{a_{n+1}}{a_n} = \frac{\sqrt{n!}}{\sqrt{(n+1)!}} = \frac{1}{\sqrt{n+1}} \longrightarrow 0.$$

The Ratio Test applies and gives absolute convergence.

(b) The series with term $(2+\cos n)/3^n$.

Since $-1 \le \cos n \le 1$, adding 2 gives $1 \le 2 + \cos n \le 3$. Consequently

$$0 \le \frac{2+\cos n}{3^n} \le \frac{3}{3^n}.$$

The comparison series is geometric, with positive constant factor 3 and ratio $1/3 < 1$. It converges, so the Comparison Test proves absolute convergence of the given nonnegative series.

(c) The series with term $1/(2^n + n)$.

We have $2^n + n \ge 2^n > 0$. Reciprocation yields

$$0 < \frac{1}{2^n + n} \le \frac{1}{2^n}.$$

The upper comparison is a geometric series of ratio $1/2$, so the requested series converges absolutely.

(d) The series with term $(1/2)^n(50 + 2/n)$.

For $n \ge 1$, the inequality $0 < 2/n \le 2$ implies $0 < 50 + 2/n \le 52$. Multiply by the positive number 2^{-n}:

$$0 < \left(\frac{1}{2}\right)^n \left(50 + \frac{2}{n}\right) \le \frac{52}{2^n}.$$

The series on the right is a constant multiple of a convergent geometric series. The Comparison Test therefore gives absolute convergence. In particular, the expression in parentheses is $50 + 2/n$, with n in the denominator of its second term.

(e) The series with term $\sin(n\pi/9)$.

Consider the strictly increasing indices $n_k = 18k + 3$, $k \in \mathbb{N}$. The corresponding terms are

$$\sin\left(\frac{(18k+3)\pi}{9}\right) = \sin(2k\pi + \pi/3) = \sin(\pi/3) = \frac{\sqrt{3}}{2}.$$

This nonzero constant subsequence prevents the whole term sequence from tending to zero: every subsequence of a sequence tending to zero would also tend to zero. Thus Corollary 14.5 proves that the series diverges. A term test does not require the terms to be nonnegative or the partial sums to tend to $+\infty$.

(f) The series with term $100^n/n!$.

For $a_n = 100^n/n! > 0$, factorial cancellation gives

$$\frac{a_{n+1}}{a_n} = \frac{100}{n+1} \longrightarrow 0 < 1.$$

The Ratio Test gives absolute convergence, despite the relatively large fixed base 100 in the numerator.

Conclusion. Every series in this exercise except (e) converges absolutely. Part (e) diverges because its terms do not tend to zero.

14.4 Exercise 14.4: Oscillating denominators, telescoping roots, and factorials

What we must show. Classify the three series, keeping track of the starting index and the placement of the powers.

(a) The series with term $1/[n+(-1)^n]^2$, for $n \geq 2$.

For these indices,

$$n + (-1)^n \geq n - 1 \geq \frac{n}{2} > 0.$$

The middle inequality is equivalent to $n \geq 2$. Because both sides are positive, squaring preserves order and taking reciprocals reverses it. We obtain

$$0 < \frac{1}{[n+(-1)^n]^2} \leq \frac{4}{n^2} \qquad (n \geq 2).$$

The comparison series converges by the p-series criterion with $p = 2$ and the constant multiple limit law. Hence the given series converges absolutely. Beginning at $n = 2$ also avoids the zero denominator that would occur at $n = 1$.

(b) The series with term $\sqrt{n+1} - \sqrt{n}$.

Each term is positive. Its Nth partial sum telescopes:

$$\begin{aligned}\sum_{n=1}^{N} (\sqrt{n+1} - \sqrt{n}) &= (\sqrt{2} - 1) + (\sqrt{3} - \sqrt{2}) + \cdots + (\sqrt{N+1} - \sqrt{N}) \\ &= \sqrt{N+1} - 1.\end{aligned}$$

All intermediate square roots cancel exactly in this finite sum. For a given $M > 0$, any $N > (M+1)^2$ makes the last expression greater than M. Thus the partial sums tend to $+\infty$ and the series diverges to $+\infty$. Individual terms tending to zero would not change this conclusion about the partial sums.

(c) The series with term $n!/n^n$.

Let $a_n = n!/n^n > 0$. The ratio simplifies as follows:

$$\begin{aligned}\frac{a_{n+1}}{a_n} &= \frac{(n+1)!}{(n+1)^{n+1}} \frac{n^n}{n!} = \frac{(n+1)n^n}{(n+1)^{n+1}} \\ &= \left(\frac{n}{n+1}\right)^n = \frac{1}{(1+1/n)^n}.\end{aligned}$$

Exercise 12.14 justified $(1+1/n)^n \to e$ with finite $e \geq 2$. The reciprocal limit law therefore gives ratio limit $1/e$, which lies in $(0, 1)$. The Ratio Test proves absolute convergence.

Conclusion. Parts (a) and (c) converge absolutely. Part (b) diverges to $+\infty$.

14.5 Exercise 14.5: Linearity of series and failure of a termwise product formula

What we must show. If $\sum a_n = A$ and $\sum b_n = B$ are finite real sums, prove the two linearity formulas and examine the proposed product formula.

Preparation: use partial sums, whose limits are given.

Write $S_N = \sum_{n=1}^{N} a_n$ and $T_N = \sum_{n=1}^{N} b_n$. The hypotheses mean $S_N \to A$ and $T_N \to B$. These are finite limits, so the sum and constant multiple laws from Section 9 apply.

(a) Addition of two series.

Finite distributivity gives

$$\sum_{n=1}^{N} (a_n + b_n) = S_N + T_N.$$

Taking limits yields $A + B$. By the definition of a series sum, this proves $\sum_{n=1}^{\infty}(a_n + b_n) = A + B$.

(b) Multiplication by a fixed real constant.

For any $k \in \mathbb{R}$ and every N,

$$\sum_{n=1}^{N} ka_n = kS_N.$$

The right side tends to kA, including when $k = 0$ or $k < 0$. Hence $\sum_{n=1}^{\infty} ka_n = kA$.

(c) Test the termwise product formula on finite series.

Let $a_1 = 1$ and $a_n = 0$ for $n \geq 2$. Let $b_2 = 1$ and $b_n = 0$ for every $n \neq 2$. Each series has sum 1, so $AB = 1$. At $n = 1$ the second factor is zero; at $n = 2$ the first factor is zero; at every other index both are zero. Thus $a_n b_n = 0$ for all n, and

$$\sum_{n=1}^{\infty} a_n b_n = 0 \neq 1 = AB.$$

The conjectured identity is false even for series with only one nonzero term each. The finite product $S_N T_N$ contains all the terms $a_i b_j$ with $1 \leq i, j \leq N$, whereas the partial sum $\sum_{n=1}^{N} a_n b_n$ includes only the terms with equal indices. The product limit law applies to $S_N T_N$, not to that shorter sum.

Conclusion. The two linearity identities hold. The termwise product identity does not follow and is false in general.

14.6 Exercise 14.6: Absolutely summable terms times a bounded sequence

What we must show. Prove (a) that $\sum a_n b_n$ converges when $\sum |a_n|$ converges and (b_n) is bounded, and (b) recover Corollary 14.7.

(a), first step: put one bound on every second factor.

Boundedness supplies a finite $M > 0$ with $|b_n| \leq M$ for every n. We may insist on $M > 0$ by enlarging a given bound, even if every b_n is zero.

(a), second step: verify the Cauchy criterion for the product series.

Let $\varepsilon > 0$. Since the nonnegative series $\sum |a_n|$ converges, its Cauchy criterion gives an index N such that

$$n \geq m > N \quad \Longrightarrow \quad \sum_{j=m}^{n} |a_j| < \frac{\varepsilon}{M}.$$

For any such m, n, the finite triangle inequality and the bound on b_j imply

$$\left|\sum_{j=m}^{n} a_j b_j\right| \leq \sum_{j=m}^{n} |a_j|\,|b_j| \leq M \sum_{j=m}^{n} |a_j| < M\frac{\varepsilon}{M} = \varepsilon.$$

This is exactly the Cauchy criterion for $\sum a_n b_n$. Theorem 14.4 therefore proves convergence. The same estimate without the outer absolute value also proves the Cauchy criterion for $\sum |a_n b_n|$, so the product series actually converges absolutely.

(b) Use a constant second sequence.

Choose $b_n = 1$ for all n. This sequence is bounded and $a_n b_n = a_n$. Part (a) then says that convergence of $\sum |a_n|$ implies convergence of $\sum a_n$, precisely Corollary 14.7.

Conclusion. Every bounded multiplier preserves absolute summability; the constant multiplier 1 gives the cited corollary.

14.7 Exercise 14.7: Powers greater than one of a summable nonnegative sequence

What we must show. If $a_n \geq 0$, $\sum a_n$ converges, and $p > 1$, prove that $\sum a_n^p$ converges.

1. A convergent series has terms tending to zero.

Corollary 14.5 gives $a_n \to 0$. Hence there is N_0 such that $0 \leq a_n < 1$ for all $n > N_0$.

2. Compare the powers only where that comparison is justified.

If $a_n = 0$, then $a_n^p = 0 \leq a_n$. If $0 < a_n \leq 1$, the positive exponent $p-1$ gives $a_n^{p-1} \leq 1$. Multiplication by $a_n > 0$ yields

$$0 \leq a_n^p = a_n a_n^{p-1} \leq a_n \qquad (n > N_0).$$

The restriction to sufficiently late indices matters: a number larger than 1 has its size increased by taking a power $p > 1$.

3. Include the finite initial part.

Let $A = \sum_{n=1}^{\infty} a_n < +\infty$ and $C = \sum_{n=1}^{N_0} a_n^p$, both nonnegative and finite. For $N > N_0$,

$$\sum_{n=1}^{N} a_n^p = C + \sum_{n=N_0+1}^{N} a_n^p \leq C + \sum_{n=N_0+1}^{N} a_n \leq C + A.$$

For $N \leq N_0$, the partial sum is at most $C \leq C + A$ as well. The partial sums of $\sum a_n^p$ are increasing because their terms are nonnegative, and the displayed inequalities bound them above. The monotone sequence theorem proves a finite limit.

Conclusion. The series $\sum a_n^p$ converges for every real $p > 1$.

14.8 Exercise 14.8: A geometric-mean comparison for two nonnegative series

What we must show. If $a_n, b_n \geq 0$ and both $\sum a_n$ and $\sum b_n$ converge, prove convergence of $\sum \sqrt{a_n b_n}$.

1. Derive the needed pointwise inequality.

The square of any real number is nonnegative. Apply this to $\sqrt{a_n} - \sqrt{b_n}$, whose square expands to

$$0 \le (\sqrt{a_n} - \sqrt{b_n})^2 = a_n + b_n - 2\sqrt{a_n b_n}.$$

Rearranging and dividing by $2 > 0$ gives

$$0 \le \sqrt{a_n b_n} \le \frac{a_n + b_n}{2} \le a_n + b_n.$$

The last inequality uses $a_n + b_n \ge 0$. The square-root identity also holds when one or both of the numbers are zero.

2. Apply the Comparison Test to a convergent upper series.

By Exercise 14.5, $\sum(a_n + b_n)$ converges, with sum equal to the sum of the two original sums. Its terms are nonnegative. The displayed pointwise inequality therefore satisfies all the hypotheses of the Comparison Test, which proves convergence of $\sum \sqrt{a_n b_n}$.

Conclusion. The geometric-mean series converges. In fact its sum is at most one half of the sum of the two given series.

14.9 Exercise 14.9: Changing finitely many terms

What we must show. Suppose $a_n = b_n$ except for finitely many indices. Prove that $\sum a_n$ and $\sum b_n$ either both converge or both diverge.

1. Place all exceptional indices before a single cutoff.

If the exceptional set is nonempty, let N_0 be its maximum. If the set is empty, take $N_0 = 1$. In either case $a_n = b_n$ for every $n > N_0$. Define the finite real constant

$$D = \sum_{n=1}^{N_0} (a_n - b_n).$$

2. Compare partial sums exactly.

For each $N \ge N_0$, all the later differences are zero, so

$$\sum_{n=1}^{N} a_n - \sum_{n=1}^{N} b_n = \sum_{n=1}^{N_0} (a_n - b_n) = D.$$

Thus the two partial-sum sequences differ by a fixed constant from that point onward.

3. Prove both directions of convergence.

If $\sum a_n = A$ is finite, the last identity gives $\sum_{n=1}^{N} b_n = \sum_{n=1}^{N} a_n - D \to A - D$, so the second series converges. Conversely, if $\sum b_n = B$ is finite, then $\sum_{n=1}^{N} a_n = \sum_{n=1}^{N} b_n + D \to B + D$. Therefore convergence of either series implies convergence of the other. If one diverges and the other converged, one of these implications would be contradicted; hence divergence is shared too.

Conclusion. Finite changes preserve convergence. They can change a finite sum by the constant D, which is why equality of sums is not asserted.

14.10 Exercise 14.10: Divergence detected by roots but not by ratios

What we must show. Give a series for which the Root Test proves divergence while neither conclusion of the Ratio Test applies.

1. Define positive terms with alternating adjacent ratios.

Take

$$a_n = 2^{\,n+(-1)^n} \qquad (n \geq 1).$$

Every term is positive, so its root and ratio are defined. The nth root is $a_n^{1/n} = 2 \cdot 2^{(-1)^n/n}$. Since $-1/n \leq (-1)^n/n \leq 1/n$ and powers of 2 preserve order,

$$2 \cdot 2^{-1/n} \leq a_n^{1/n} \leq 2 \cdot 2^{1/n}.$$

Both outer expressions tend to 2, using the root limit for the fixed positive number 2 and its reciprocal. The squeeze theorem gives $a_n^{1/n} \to 2$.

2. Apply the divergent part of the Root Test.

The root limit superior is $2 > 1$, so the Root Test proves divergence of $\sum a_n$.

3. Check both possible ratio-test inequalities.

When n is even, $(-1)^n = 1$ and $(-1)^{n+1} = -1$, giving

$$\frac{a_{n+1}}{a_n} = 2^{(n+1)-1-(n+1)} = 2^{-1} = \frac{1}{2}.$$

When n is odd, the two signs are reversed, giving

$$\frac{a_{n+1}}{a_n} = 2^{(n+1)+1-(n-1)} = 2^3 = 8.$$

Each value occurs infinitely often, and there are no other ratio values. Hence the ratio limit inferior is $1/2$ and its limit superior is 8. The convergence condition, ratio limit superior less than 1, fails. The divergence condition, ratio limit inferior greater than 1, fails as well. Thus the Ratio Test is inconclusive even though the Root Test settles the question.

Conclusion. The series $\sum 2^{n+(-1)^n}$ has the required properties: root limit 2, ratio limit inferior $1/2$, and ratio limit superior 8.

14.11 Exercise 14.11: A constant successive ratio gives a geometric series

What we must show. Suppose every $a_n \neq 0$ and $a_{n+1}/a_n = r$ is independent of n. Prove that $\sum a_n$ is geometric.

1. Convert the quotient statement to a recurrence.

Since $a_n \neq 0$, multiplication by a_n gives $a_{n+1} = ra_n$ for every n. Also $r = a_2/a_1 \neq 0$ because both a_1 and a_2 are nonzero.

2. Solve the recurrence by induction.

At $n = 1$, the formula $a_n = a_1 r^{n-1}$ reads $a_1 = a_1 r^0$, which holds. Suppose $a_n = a_1 r^{n-1}$ for some $n \geq 1$. Then

$$a_{n+1} = r a_n = r(a_1 r^{n-1}) = a_1 r^n.$$

This is the same formula with n replaced by $n + 1$, so induction proves it for every positive integer.

3. Recognize the series and its possible sum.

The series is therefore

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} a_1 r^{n-1} = a_1 + a_1 r + a_1 r^2 + \cdots,$$

which is the definition of a geometric series with first term a_1 and common ratio r. It converges when $|r| < 1$ and then has sum $a_1/(1 - r)$. If $|r| \geq 1$, the nonzero first term ensures that the terms do not tend to zero, so it diverges.

Conclusion. A constant successive ratio forces the exact geometric form $a_n = a_1 r^{n-1}$.

14.12 Exercise 14.12: Selecting an absolutely summable subsequence

What we must show. From $\liminf |a_n| = 0$, construct strictly increasing indices n_k for which $\sum_{k=1}^{\infty} a_{n_k}$ converges.

1. Arbitrarily small absolute values occur after every finite prefix.

Fix a number $\eta > 0$ and an index cutoff $N \geq 0$. If no $n > N$ satisfied $|a_n| < \eta$, every term beyond N would have $|a_n| \geq \eta$. All later tail infima would then be at least η, which would imply $\liminf |a_n| \geq \eta > 0$, a contradiction. Thus such an index always exists.

2. Select a geometric error bound at each stage.

Put $n_0 = 0$. After n_{k-1} has been chosen, use Step 1 with $N = n_{k-1}$ and $\eta = 2^{-k}$ to choose

$$n_k > n_{k-1}, \qquad |a_{n_k}| < 2^{-k}.$$

For definiteness, choose the least eligible index. Existence is provided by Step 1, and the least positive integer principle makes the recursive choice well defined. Strict increase of the indices ensures that (a_{n_k}) is a subsequence of the original sequence.

3. Compare the absolute-value series to a geometric series.

The geometric series satisfies $\sum_{k=1}^{\infty} 2^{-k} = 1$. Its terms dominate $|a_{n_k}| \geq 0$, so the Comparison Test gives convergence of $\sum |a_{n_k}|$. Absolute convergence implies convergence of $\sum a_{n_k}$ by Corollary 14.7. The geometric bound is essential to this argument: merely choosing a subsequence whose terms tend to zero would not by itself prove that its series converges.

Conclusion. There is an absolutely convergent subsequence series, obtained by imposing $|a_{n_k}| < 2^{-k}$ at stage k.

14.13 Exercise 14.13: Four exact sums from geometric and telescoping identities

What we must show. Calculate the two geometric sums in part (a), prove the identities in parts (b) and (c), and deduce the sum in part (d).

(a) The ratios $2/3$ and $-2/3$, starting at $n = 1$.

For any $|r| < 1$, shifting the index of the geometric series gives

$$\sum_{n=1}^{\infty} r^n = r \sum_{j=0}^{\infty} r^j = \frac{r}{1-r}.$$

Both ratios here have absolute value $2/3 < 1$. Substitution yields

$$\sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^n = \frac{2/3}{1-2/3} = 2, \qquad \sum_{n=1}^{\infty} \left(-\frac{2}{3}\right)^n = \frac{-2/3}{1+2/3} = -\frac{2}{5}.$$

The negative sign in the second first term matters because its sum begins at exponent 1, rather than exponent 0.

(b) The series with term $1/[n(n+1)]$.

The finite algebraic identity

$$\frac{1}{n} - \frac{1}{n+1} = \frac{n+1-n}{n(n+1)} = \frac{1}{n(n+1)}$$

gives, for every $N \geq 1$,

$$\begin{aligned} \sum_{n=1}^{N} \frac{1}{n(n+1)} &= (1 - \tfrac{1}{2}) + (\tfrac{1}{2} - \tfrac{1}{3}) + \cdots + (\tfrac{1}{N} - \tfrac{1}{N+1}) \\ &= 1 - \frac{1}{N+1}. \end{aligned}$$

Taking the limit $N \to \infty$ gives the finite sum 1.

(c) The series with term $(n-1)/2^{n+1}$.

Put the right side of the following identity over a common denominator to verify it:

$$\frac{n}{2^n} - \frac{n+1}{2^{n+1}} = \frac{2n-(n+1)}{2^{n+1}} = \frac{n-1}{2^{n+1}}.$$

The first term of the series, at $n = 1$, is zero. Summing the identity to N causes all intervening terms to cancel:

$$\sum_{n=1}^{N} \frac{n-1}{2^{n+1}} = \frac{1}{2} - \frac{N+1}{2^{N+1}}.$$

To justify the last term's limit explicitly, set $u_N = (N+1)/2^{N+1} > 0$. For $N \geq 1$,

$$\frac{u_{N+1}}{u_N} = \frac{N+2}{2(N+1)} \leq \frac{3}{4},$$

because the last inequality is equivalent to $N \geq 1$. Induction gives $0 < u_N \leq u_1(3/4)^{N-1}$, whose upper bound tends to zero. Hence $u_N \to 0$ by the squeeze theorem, and

$$\sum_{n=1}^{\infty} \frac{n-1}{2^{n+1}} = \frac{1}{2}.$$

(d) Deduce the sum with term $n/2^n$.

For each index,

$$\frac{n}{2^n} = 2\frac{n-1}{2^{n+1}} + \frac{1}{2^n}.$$

The first series on the right converges by part (c), and the second is geometric with sum 1. The linearity formulas of Exercise 14.5 therefore apply and give

$$\sum_{n=1}^{\infty} \frac{n}{2^n} = 2\left(\frac{1}{2}\right) + 1 = 2.$$

Conclusion. The requested sums are, in order, 2, $-2/5$, 1, $1/2$, and 2. All telescoping cancellations were made in finite partial sums before taking a limit.

14.14 Exercise 14.14: A blockwise lower comparison for the harmonic series

What we must show. Prove divergence of $\sum_{n=1}^{\infty} 1/n$ by comparing it with the displayed repeated-block sequence.

1. Express the displayed sequence by exact index ranges.

For each integer $j \geq 1$, the block

$$2^{j-1} \leq n \leq 2^j - 1$$

contains $2^j - 1 - 2^{j-1} + 1 = 2^{j-1}$ indices. On that block define $a_n = 2^{-j}$. Thus the first block is the single term $1/2$; the second contains two terms $1/4$; the third contains four terms $1/8$; and the fourth contains eight terms $1/16$. These blocks exactly reproduce the sequence in the question.

2. Every complete block contributes the same positive amount.

For the block labeled j, its sum is

$$2^{j-1} \cdot 2^{-j} = \frac{1}{2}.$$

The comparison series in the exercise begins at $n = 2$, so it omits the first block and retains all blocks with $j \geq 2$. At the end of block $J \geq 2$ its partial sum is

$$\sum_{n=2}^{2^J-1} a_n = \sum_{j=2}^{J} \frac{1}{2} = \frac{J-1}{2}.$$

These values tend to $+\infty$ as J grows. All intermediate partial sums are increasing, since $a_n > 0$. Therefore the entire comparison series, not only these block-end partial sums, diverges to $+\infty$.

3. Compare each harmonic term with its block value.

Given $n \geq 2$, there is a unique block index $j \geq 2$ with $2^{j-1} \leq n \leq 2^j - 1$. Since $n < 2^j$ and both are positive,

$$\frac{1}{n} > \frac{1}{2^j} = a_n.$$

The Comparison Test now shows that $\sum_{n=2}^{\infty} 1/n$ diverges to $+\infty$. Adding the first harmonic term, which is 1, preserves this divergence.

Conclusion. The harmonic series diverges to $+\infty$ because it dominates a positive series whose every complete block adds $1/2$.

15 Alternating series and integral tests

In this section, log means the natural logarithm. A series is *conditionally convergent* if it converges but its series of absolute values diverges. Changing or removing finitely many terms does not change convergence, by Exercise 14.9. We use this fact when a comparison or monotonicity estimate holds only after a fixed index.

The alternating-series theorem requires nonnegative magnitudes that decrease to zero. An alternating sign by itself does not guarantee convergence. An integral test requires a nonnegative decreasing function matching the terms; the precise statement and proof are given in Exercise 15.8 below. As in the textbook's Section 15, the usual elementary integral and derivative formulas are available.

15.1 Exercise 15.1: Two alternating series with very different magnitudes

What we must show. Decide whether the series in parts (a) and (b) converge, and justify each answer.

Plan. Check the magnitudes before applying the alternating-series theorem. In particular, a convergent series must have terms tending to zero, regardless of their signs.

(a) The series $\sum_{n=1}^{\infty}(-1)^n/n$.

Put $b_n = 1/n$. For every positive integer n, we have $b_n > 0$. Also $n+1 > n > 0$, so taking reciprocals reverses the inequality:

$$b_{n+1} = \frac{1}{n+1} < \frac{1}{n} = b_n.$$

Finally, $b_n \to 0$. Indeed, for $\varepsilon > 0$, choose an integer $N > 1/\varepsilon$. If $n > N$, then $|b_n - 0| = 1/n < \varepsilon$.

Thus $\sum_{n=1}^{\infty}(-1)^{n+1}b_n$ converges by the alternating-series theorem. Multiplication by -1 preserves convergence, by Exercise 14.5(b), and gives the series requested here. Its series of absolute values is $\sum_{n=1}^{\infty} 1/n$, the divergent harmonic series. Hence the requested series converges conditionally.

(b) The series $\sum_{n=1}^{\infty}(-1)^n n!/2^n$.

Let $c_n = n!/2^n > 0$. Cancellation of the common factorial and power gives

$$\frac{c_{n+1}}{c_n} = \frac{(n+1)!}{2^{n+1}}\frac{2^n}{n!} = \frac{n+1}{2}.$$

For $n \geq 3$, this ratio is at least 2. Repeated multiplication therefore gives, for $n \geq 3$,

$$c_n \geq c_3\, 2^{n-3} = \frac{3}{4}\, 2^{n-3}.$$

The right side tends to $+\infty$. In particular c_n does not tend to zero. If the signed terms $(-1)^n c_n$ tended to zero, their absolute values c_n would tend to zero too, a contradiction. The necessary term condition, Corollary 14.5, now proves divergence. Alternation does not compensate for these growing magnitudes.

Conclusion. Part (a) converges conditionally. Part (b) diverges because its terms do not tend to zero.

15.2 Exercise 15.2: Powers of periodically sampled sine values

What we must show. Determine convergence of

$$\text{(a)} \quad \sum_{n=1}^{\infty} \left[\sin\left(\frac{n\pi}{6}\right)\right]^n, \qquad \text{(b)} \quad \sum_{n=1}^{\infty} \left[\sin\left(\frac{n\pi}{7}\right)\right]^n.$$

The exponent n applies to the whole sine value in each case.

Plan. In part (a), the sampled sine values reach 1 infinitely often. In part (b), there are only finitely many sampled values, and every one has absolute value strictly less than 1.

(a) Exhibit terms that stay equal to 1.

For every integer $k \geq 0$, choose $n_k = 12k + 3$. These are strictly increasing positive indices, and

$$\frac{n_k\pi}{6} = 2k\pi + \frac{\pi}{2}.$$

Sine has period 2π and $\sin(\pi/2) = 1$, so

$$\left[\sin\left(\frac{n_k\pi}{6}\right)\right]^{n_k} = 1^{n_k} = 1.$$

The terms of a convergent series must tend to zero, and every subsequence of a sequence tending to zero must also tend to zero. This constant subsequence contradicts that requirement. Therefore the series in part (a) diverges.

(b) Bound all the absolute sine values by one number below 1.

The values $\sin(n\pi/7)$ repeat when n is increased by 14, because the angle then increases by 2π. Thus all their absolute values occur in the finite list

$$\left\{\left|\sin\left(\frac{r\pi}{7}\right)\right| : r = 1, 2, \ldots, 14\right\}.$$

None of these absolute values equals 1. Otherwise the angle $r\pi/7$ would have the form $\pi/2 + \ell\pi$ for some integer ℓ. After division by π and multiplication by 14, this would give $2r = 7(2\ell + 1)$. The left side is even and the right side is odd, which is impossible.

Let c be the maximum of that finite list. Every entry is less than 1, so its attained maximum satisfies $c < 1$ as well; also $c \geq 0$. In fact $c > 0$, since $\sin(\pi/7) > 0$. For every $n \geq 1$,

$$\left|\left[\sin\left(\frac{n\pi}{7}\right)\right]^n\right| = \left|\sin\left(\frac{n\pi}{7}\right)\right|^n \leq c^n.$$

The inequality is preserved when taking the positive integer power because both sides are nonnegative. The geometric series $\sum_{n=1}^{\infty} c^n$ converges, since $0 < c < 1$. Comparison proves that the series of absolute values converges. Hence the series in part (b) converges absolutely.

Conclusion. Part (a) diverges by the necessary term condition. Part (b) converges absolutely by comparison with a geometric series.

15.3 Exercise 15.3: The logarithmic refinement of the harmonic series

What we must show. For a real parameter p, prove

$$\sum_{n=2}^{\infty} \frac{1}{n(\log n)^p} \text{ converges} \quad \Longleftrightarrow \quad p > 1.$$

Plan. For positive p, use the integral test with the substitution $u = \log x$. For nonpositive p, compare directly with $1/n$.

1. Check positivity and the case $p \leq 0$.

For $n \geq 2$, $\log n > 0$, so every real power $(\log n)^p$ is defined and positive. Choose an integer $N > e$. For $n \geq N$, $\log n > 1$. If $p \leq 0$, then $(\log n)^p \leq 1$, and therefore

$$\frac{1}{n(\log n)^p} \geq \frac{1}{n}.$$

The harmonic series diverges. Comparison shows that the series under consideration diverges to $+\infty$ for $p \leq 0$.

2. For $p > 0$, verify the hypotheses of the integral test.

Define

$$f_p(x) = \frac{1}{x(\log x)^p} \qquad (x \geq 2).$$

The function is positive and continuous on this interval. Differentiating the product $x^{-1}(\log x)^{-p}$ gives

$$\begin{aligned} f_p'(x) &= -x^{-2}(\log x)^{-p} - px^{-2}(\log x)^{-p-1} \\ &= -\frac{\log x + p}{x^2(\log x)^{p+1}} < 0. \end{aligned}$$

The denominator is positive, and the numerator $\log x + p$ is positive because $x \geq 2$ and $p > 0$. Thus the function decreases and matches the series terms at every integer $n \geq 2$.

3. Evaluate the integrals and distinguish the parameter cases.

For $R \geq 2$, put $u = \log x$, so $du = dx/x$. Then

$$\int_2^R \frac{dx}{x(\log x)^p} = \int_{\log 2}^{\log R} u^{-p}\, du.$$

If $p \neq 1$, the power rule gives

$$\int_2^R \frac{dx}{x(\log x)^p} = \frac{(\log R)^{1-p} - (\log 2)^{1-p}}{1-p}.$$

For $p > 1$, one has $1 - p < 0$, so $(\log R)^{1-p} \to 0$ as $R \to \infty$. The integral tends to the finite positive number $(\log 2)^{1-p}/(p-1)$. The integral test proves convergence.

For $0 < p < 1$, one has $1 - p > 0$, so $(\log R)^{1-p} \to +\infty$. The denominator $1 - p$ is positive, and the integral tends to $+\infty$. The integral test proves divergence.

For $p = 1$, integration instead gives

$$\int_2^R \frac{dx}{x \log x} = \log(\log R) - \log(\log 2) \longrightarrow +\infty.$$

The arguments of both outer logarithms are positive, even though $\log(\log 2)$ itself is negative. This case also diverges.

Conclusion. Exactly the parameters $p > 1$ give convergence. For every $p \leq 1$, the nonnegative series diverges to $+\infty$.

15.4 Exercise 15.4: Four series involving logarithms

What we must show. Determine convergence of each of the four requested series.

Plan. Compare the first two series with the harmonic series. For the other two, check the integral-test hypotheses and compute the relevant integrals.

(a) The series $\sum_{n=2}^{\infty} 1/(\sqrt{n} \log n)$.

We first justify the eventual inequality $\log x \leq \sqrt{x}$. Let $g(x) = \sqrt{x} - \log x$ for $x > 0$. Its derivative is

$$g'(x) = \frac{1}{2\sqrt{x}} - \frac{1}{x} = \frac{\sqrt{x} - 2}{2x} \geq 0 \qquad (x \geq 4).$$

Also $\log 2 = \int_1^2 dt/t < 1$, since $1/t < 1$ on $(1, 2]$. Thus $g(4) = 2 - \log 4 = 2 - 2\log 2 > 0$. As g is nondecreasing on $[4, \infty)$, we have $\log x < \sqrt{x}$ there.

For integers $n \geq 4$, both logarithms and square roots are positive, and the inequality just proved gives

$$\sqrt{n} \log n \leq n, \qquad \frac{1}{\sqrt{n} \log n} \geq \frac{1}{n}.$$

Taking reciprocals reverses the positive-denominator inequality. Comparison with the divergent harmonic series proves divergence to $+\infty$.

(b) The series $\sum_{n=2}^{\infty} (\log n)/n$.

Choose an integer $N > e$. If $n \geq N$, monotonicity of the logarithm gives $\log n > \log e = 1$. Consequently

$$\frac{\log n}{n} \geq \frac{1}{n}.$$

All terms are positive for $n \geq 2$. Comparison with the harmonic series, with the first finitely many indices ignored, proves divergence to $+\infty$.

(c) The series $\sum_{n=4}^{\infty} 1/[n(\log n)(\log \log n)]$.

For $x \geq 4$, we have $\log x > 1$. One direct check at the left endpoint is

$$\log 2 = \int_1^2 \frac{dt}{t} > \frac{1}{2}, \qquad \log 4 = 2 \log 2 > 1.$$

Therefore $\log \log x > 0$ on this whole interval. The three factors x, $\log x$, and $\log \log x$ are positive, continuous, and increasing. Their product is increasing, so its positive reciprocal

$$f(x) = \frac{1}{x(\log x)(\log \log x)}$$

is decreasing and continuous on $[4, \infty)$.

Two substitutions, first $u = \log x$ and then $v = \log u$, give

$$\begin{aligned}\int_4^R \frac{dx}{x(\log x)(\log\log x)} &= \int_{\log 4}^{\log R} \frac{du}{u \log u} \\ &= \int_{\log\log 4}^{\log\log R} \frac{dv}{v} \\ &= \log(\log(\log R)) - \log(\log(\log 4)).\end{aligned}$$

Every argument of a logarithm in these formulas is positive for $R \geq 4$: in particular $\log\log 4 > 0$. As $R \to \infty$, the first triple logarithm tends to $+\infty$. The integral test proves that this series diverges to $+\infty$.

(d) The series $\sum_{n=2}^{\infty} (\log n)/n^2$.

Let $h(x) = (\log x)/x^2$ for $x \geq 2$. This function is positive and continuous. Differentiation gives

$$h'(x) = \frac{1 - 2\log x}{x^3} < 0 \qquad (x \geq 2),$$

because $\log x \geq \log 2 > 1/2$, as established in part (c). Thus the integral test applies. For integration by parts, take $u = \log x$ and $dv = x^{-2}\,dx$. Then $du = dx/x$ and $v = -1/x$, so

$$\begin{aligned}\int_2^R \frac{\log x}{x^2}\,dx &= \left[-\frac{\log x}{x}\right]_2^R + \int_2^R \frac{dx}{x^2} \\ &= \frac{\log 2 + 1}{2} - \frac{\log R + 1}{R}.\end{aligned}$$

Part (a) gives $0 \leq (\log R)/R \leq 1/\sqrt{R}$ for $R \geq 4$. The right side tends to zero, as does $1/R$. Hence the integral tends to $(\log 2 + 1)/2$, a finite number. The integral test proves convergence of the series.

Conclusion. Parts (a), (b), and (c) diverge to $+\infty$. Part (d) converges.

15.5 Exercise 15.5: Why a direct comparison does not establish all positive powers

What we must show. Explain why the text used an integral argument to prove the convergence of $\sum 1/n^p$ for every $p > 1$.

Plan. A comparison proof needs an independently known convergent upper series. It cannot assume a nearby case of the very theorem it is trying to establish.

1. Check which direction of comparison is useful.

Knowing that $\sum 1/n^2$ converges immediately handles $p \geq 2$, since $0 \leq 1/n^p \leq 1/n^2$ for $n \geq 1$. For $1 < p < 2$, however, the inequality reverses:

$$\frac{1}{n^p} \geq \frac{1}{n^2}.$$

Being larger than the terms of a convergent nonnegative series does not establish either convergence or divergence. Comparison with $1/n$ also fails to settle the question: for $p > 1$ the terms $1/n^p$ are smaller than the terms of that divergent series.

2. Explain the missing starting exponent.

If convergence were already known at some $p_0 > 1$, comparison would prove it for every $p \geq p_0$. It would leave the interval $1 < p < p_0$ unresolved. There is no smallest real exponent greater than 1: between 1 and p_0 lies $(1 + p_0)/2$. Thus no single known case $p_0 > 1$ settles the whole range by this direct comparison. Choosing an unproved exponent q with $1 < q < p$ merely replaces the original convergence problem with another case of the same problem. The integral calculation supplies an independent argument for every $p > 1$.

3. Comparison can still work after grouping the terms.

The explanation above does not mean a comparison proof is impossible. For example, fix $p > 1$ and group the indices into blocks

$$2^j \leq n < 2^{j+1} \qquad (j = 0, 1, 2, \ldots).$$

There are exactly 2^j integers in this block, and each satisfies $n^{-p} \leq 2^{-jp}$. Hence the sum of the block is at most

$$2^j 2^{-jp} = 2^{j(1-p)}.$$

The bounding block sums form a geometric series with ratio $2^{1-p} < 1$. Every original partial sum is bounded above by the sum of this geometric series, since its terms are nonnegative and its indices occupy finitely many blocks. The original partial sums are increasing and bounded above, so they converge by Theorem 10.2. This is a valid alternative proof using an extra grouping argument.

Conclusion. The integral proof avoids relying on an unproved nearby p-series. A plain comparison with one known exponent does not cover all $p > 1$; a geometric comparison of grouped blocks does.

15.6 Exercise 15.6: What squaring the terms can change

What we must show. Supply the examples in parts (a) and (c), and justify the nonnegative-term assertion in part (b).

(a) A divergent series whose squared-term series converges.

Take $a_n = 1/n$. Then $\sum a_n$ is the divergent harmonic series, whereas

$$\sum_{n=1}^{\infty} a_n^2 = \sum_{n=1}^{\infty} \frac{1}{n^2}$$

converges by the p-series theorem, with $p = 2 > 1$. These two facts verify both required properties of the example.

(b) Squaring preserves convergence when the original terms are nonnegative.

Suppose $a_n \geq 0$ and $\sum a_n$ converges. Corollary 14.5 gives $a_n \to 0$. There is therefore an integer N with $0 \leq a_n < 1$ whenever $n > N$. Multiplication of $a_n < 1$ by the nonnegative number a_n gives

$$0 \leq a_n^2 \leq a_n \qquad (n > N).$$

The comparison test proves convergence of $\sum_{n>N} a_n^2$. Adding its finitely many omitted real terms preserves convergence, so the whole squared-term series converges. This is also Exercise 14.7 with the power $p = 2$.

(c) A convergent signed series whose squared-term series diverges.

Take

$$a_n = \frac{(-1)^{n+1}}{\sqrt{n}}.$$

Its magnitudes $b_n = 1/\sqrt{n}$ are positive and decrease, because the positive square root increases with its argument and taking reciprocals reverses order. They tend to zero: given $\varepsilon > 0$, choose $N > 1/\varepsilon^2$; then $n > N$ implies $1/\sqrt{n} < \varepsilon$. The alternating-series theorem proves that $\sum a_n$ converges. However,

$$a_n^2 = \frac{[(-1)^{n+1}]^2}{(\sqrt{n})^2} = \frac{1}{n},$$

so $\sum a_n^2$ is the divergent harmonic series.

Conclusion. Use $a_n = 1/n$ for part (a) and $a_n = (-1)^{n+1}/\sqrt{n}$ for part (c). Part (b) follows from the eventual bound $0 \le a_n^2 \le a_n$.

15.7 Exercise 15.7: A necessary decay rate for decreasing summable terms

What we must show. Prove that if (a_n) is a decreasing real sequence and $\sum a_n$ converges, then $na_n \to 0$. Then apply this conclusion to the harmonic series.

Plan. First deduce that every a_n is nonnegative. A late block of roughly $n/2$ terms is small by the Cauchy criterion, yet every term in that block is at least a_n.

(a), first: derive the sign condition from the actual hypotheses.

Since $\sum a_n$ converges, Corollary 14.5 gives $a_n \to 0$. The word decreasing means $a_{n+1} \le a_n$; strict decrease is not required. If $a_j < 0$ at any index j, then $a_n \le a_j < 0$ for all $n \ge j$. Such a tail cannot tend to zero: with $\varepsilon = -a_j/2 > 0$, its terms satisfy $|a_n| \ge -a_j > \varepsilon$. This contradicts $a_n \to 0$. Thus $a_n \ge 0$ for every index n.

(a), second: use a tail block with an explicit number of terms.

Fix $\varepsilon > 0$. By the Cauchy criterion for the convergent series, choose an integer $N \ge 1$ such that

$$q \ge m > N \quad \Longrightarrow \quad \left| \sum_{j=m}^{q} a_j \right| < \frac{\varepsilon}{2}.$$

For any integer $n > 2N$, set $k = \lfloor n/2 \rfloor$. Here $\lfloor x \rfloor$ is the greatest integer at most x. We have $k \ge N$, so the block starting at $k+1$ begins after N. It contains exactly $n - k$ terms, and $n - k \ge n/2$. For $k + 1 \le j \le n$, decreasingness gives $a_j \ge a_n$. Consequently

$$0 \le \frac{n}{2} a_n \le (n-k) a_n \le \sum_{j=k+1}^{n} a_j = \left| \sum_{j=k+1}^{n} a_j \right| < \frac{\varepsilon}{2}.$$

The equality with the absolute value uses nonnegativity of the terms. Multiplying the resulting strict upper bound by 2 gives $0 \le na_n < \varepsilon$ whenever $n > 2N$. This verifies the definition of $na_n \to 0$.

(b) Apply the necessary condition to $a_n = 1/n$.

The sequence $1/n$ is decreasing. If the harmonic series converged, part (a) would imply

$$n\left(\frac{1}{n}\right) \longrightarrow 0.$$

The expression on the left equals 1 for every n, so its limit is 1, a contradiction. Therefore the harmonic series diverges. Since its terms are nonnegative, its partial sums are increasing; having no finite limit, they tend to $+\infty$ by the monotone sequence theorem for unbounded increasing sequences.

Conclusion. Every decreasing real sequence with a convergent series satisfies $na_n \to 0$. The constant value $n(1/n) = 1$ therefore provides another proof of divergence of the harmonic series.

15.8 Exercise 15.8: A general integral test and its error bounds

What we must show. State and prove a precise integral test of the type described in Discussion 15.2.

Plan. Compare each unit-width rectangle with the integral below and above it. Nonnegativity makes both the partial sums and the integrals increasing, so a finite upper bound implies convergence.

1. State the assumptions and the conclusion.

Let $m \geq 1$ be an integer, and let $f : [m, \infty) \to [0, \infty)$ be continuous and nonincreasing. Set $a_n = f(n)$ for every integer $n \geq m$. Define the improper integral by

$$\int_m^\infty f(x)\,dx = \lim_{R\to\infty} \int_m^R f(x)\,dx,$$

where the value $+\infty$ is allowed. Then

$$\sum_{n=m}^{\infty} a_n \text{ converges} \quad \Longleftrightarrow \quad \int_m^\infty f(x)\,dx < +\infty.$$

If either side is infinite, both the integral and the series diverge to $+\infty$. If they converge and $R_N = \sum_{n=N+1}^{\infty} a_n$ for $N \geq m$, then additionally

$$\int_{N+1}^\infty f(x)\,dx \leq R_N \leq \int_N^\infty f(x)\,dx.$$

Continuity is a sufficient condition ensuring that all the finite integrals used here exist. The proof also works whenever those integrals exist, with the same nonnegativity and monotonicity assumptions.

2. Establish the unit-interval inequalities.

Fix an integer $k \geq m$. If $k \leq x \leq k+1$, nonincreasingness gives $f(k+1) \leq f(x) \leq f(k)$. Integrating these constant bounds over the interval of length 1 yields

$$f(k+1) \leq \int_k^{k+1} f(x)\,dx \leq f(k).$$

In particular, summing the right-hand inequalities for $k = m, m+1, \ldots, N$ gives

$$\int_m^{N+1} f(x)\,dx \leq \sum_{k=m}^{N} f(k).$$

For the opposite comparison, use $f(k) \le \int_{k-1}^{k} f(x)\,dx$ for $k \ge m+1$. Keep the first term $f(m)$ separate and sum to obtain

$$\sum_{k=m}^{N} f(k) \le f(m) + \int_{m}^{N} f(x)\,dx.$$

At $N = m$, the second sum has no remaining terms and the integral is zero, so this estimate still holds. Combining the two bounds, with $S_N = \sum_{k=m}^{N} f(k)$, gives

$$\boxed{\int_{m}^{N+1} f(x)\,dx \le S_N \le f(m) + \int_{m}^{N} f(x)\,dx.}$$

3. A finite integral bounds the partial sums.

Suppose the improper integral has finite value I. Since $f \ge 0$, one has $\int_m^N f \le I$. Therefore $0 \le S_N \le f(m) + I$ for every $N \ge m$. Also $S_{N+1} - S_N = f(N+1) \ge 0$, so (S_N) is increasing. The monotone sequence theorem gives a finite limit of (S_N), which is precisely convergence of the series.

4. A convergent series bounds all the finite integrals.

Conversely, suppose $S_N \to S < +\infty$. The nonnegative terms imply $S_N \le S$. If $R \ge m$ is any real number, choose an integer $N \ge m$ with $R \le N+1$. Nonnegativity and the first rectangle estimate give

$$0 \le \int_{m}^{R} f(x)\,dx \le \int_{m}^{N+1} f(x)\,dx \le S_N \le S.$$

Thus all the integrals are bounded above by S, not only those with integer endpoints. They increase with R, again because $f \ge 0$. Their supremum is finite, and their limit as $R \to \infty$ equals that supremum: given a positive tolerance, choose an endpoint whose integral is within that tolerance of the supremum; every larger endpoint gives an integral between that value and the supremum. Hence the improper integral converges.

5. Explain divergence and derive the tail estimates.

If the integral tends to $+\infty$, the lower bound $S_N \ge \int_m^{N+1} f$ forces $S_N \to +\infty$. If instead $S_N \to +\infty$, the upper rectangle bound rearranges to $\int_m^N f \ge S_N - f(m)$, which forces the integral to $+\infty$ as well. Nonnegative increasing sums and integrals have no other type of divergence.

Finally, in the convergent case, apply the same unit-interval comparisons to the terms with indices $N+1, \dots, M$, where $M > N \ge m$:

$$\int_{N+1}^{M+1} f(x)\,dx \le \sum_{k=N+1}^{M} f(k) \le \int_{N}^{M} f(x)\,dx.$$

Letting $M \to \infty$ gives the stated remainder bounds. These limits are finite since the whole integral and series converge.

The same test applies if the assumptions hold only after some integer index: start at that index and use Exercise 14.9 to restore the finite initial segment.

Conclusion. For a continuous nonnegative nonincreasing function matching the terms, the series converges exactly when its improper integral is finite. The two neighboring tail integrals bound the omitted tail of the series.

16 Decimal expansions of real numbers

A bar marks exactly the digits that repeat. Thus $0.0\overline{2} = 0.02222\ldots$, whereas $0.\overline{02} = 0.020202\ldots$. A terminating decimal is understood to continue with zeros. For a decimal $K.d_1d_2\cdots$, its value means the convergent series

$$K + \sum_{j=1}^{\infty} d_j 10^{-j}, \qquad 0 \le d_j \le 9.$$

Convergence follows by comparison with the geometric series $\sum_{j=1}^{\infty} 9 \cdot 10^{-j} = 1$. Theorem 16.3 describes the only possible ambiguity: a positive terminating decimal also has a representation ending in nines.

16.1 Exercise 16.1: Two expansions for eleven fourths

What we must show. Verify both decimal representations in part (a), and identify which one long division produces in part (b).

(a) Evaluate the expansion ending in zeros.

The expression $2.75\overline{0}$ means

$$2 + \frac{7}{10} + \frac{5}{100} = \frac{200 + 70 + 5}{100} = \frac{275}{100} = \frac{11}{4}.$$

Every remaining term is zero, so there is no additional contribution.

(a) Evaluate the expansion ending in nines.

In $2.74\overline{9}$ the first nine is in the third decimal place. The sum of the nines is

$$\sum_{j=3}^{\infty} 9 \cdot 10^{-j} = 9 \cdot 10^{-3} \sum_{k=0}^{\infty} (10^{-1})^k = \frac{9 \cdot 10^{-3}}{1 - 10^{-1}} = \frac{1}{100}.$$

Consequently

$$2.74\overline{9} = 2 + \frac{7}{10} + \frac{4}{100} + \frac{1}{100} = 2 + \frac{7}{10} + \frac{5}{100} = \frac{11}{4}.$$

The extra tail of nines supplies exactly the missing hundredth.

(b) Carry out the long division.

First $11 = 4 \cdot 2 + 3$, so the integer part is 2 and the initial remainder is 3. Multiplying the remainder by ten gives

$$30 = 4 \cdot 7 + 2, \qquad 20 = 4 \cdot 5 + 0.$$

The successive decimal digits are therefore 7 and 5, after which the remainder is zero. All subsequent divisions have $10 \cdot 0 = 4 \cdot 0 + 0$, so all subsequent digits are zero.

Conclusion. Both $2.74\overline{9}$ and $2.75\overline{0}$ equal $11/4$. The long division algorithm produces $2.75\overline{0}$.

16.2 Exercise 16.2: Trading a terminating tail for a tail of nines

What we must show. Verify the two identities in the opening paragraph of the proof of Theorem 16.3: the identity for a positive integer, and the identity for a decimal with a last nonzero digit.

First calculate a general tail of nines.

For any integer $r \geq 0$, a geometric series gives

$$\sum_{j=r+1}^{\infty} 9 \cdot 10^{-j} = \frac{9 \cdot 10^{-(r+1)}}{1 - 10^{-1}} = 10^{-r}.$$

In particular, when $r = 0$ this tail has value 1.

A positive integer has the asserted alternative representation.

Let K be a positive integer. Then $K - 1$ is a nonnegative integer, so it is an allowed integer part for a nonnegative decimal. The tail formula with $r = 0$ gives

$$(K-1).\overline{9} = (K-1) + \sum_{j=1}^{\infty} 9 \cdot 10^{-j} = (K-1) + 1 = K = K.\overline{0}.$$

These digit strings are distinct even though their values agree.

A last nonzero digit can be reduced by one.

Suppose

$$x = K.d_1 \cdots d_{r-1} d_r \overline{0}, \qquad r \geq 1, \qquad d_r \in \{1, \ldots, 9\}.$$

The condition $d_r \neq 0$ makes $d_r - 1$ an allowed digit. The value of the proposed other expansion is

$$\begin{aligned} &K + \sum_{j=1}^{r-1} d_j 10^{-j} + (d_r - 1)10^{-r} + \sum_{j=r+1}^{\infty} 9 \cdot 10^{-j} \\ &\quad = K + \sum_{j=1}^{r-1} d_j 10^{-j} + (d_r - 1)10^{-r} + 10^{-r} \\ &\quad = K + \sum_{j=1}^{r-1} d_j 10^{-j} + d_r 10^{-r} = x. \end{aligned}$$

Thus decreasing the last nonzero digit by one and replacing the following zeros by nines preserves the represented number.

Check the zero case separately.

The first identity requires $K > 0$, and the second requires a last nonzero digit. Neither condition holds for $0.\overline{0}$. Indeed, if a nonnegative decimal represents zero, its nonnegative integer part must be zero. Every digit must then be zero too: a positive digit d_j would contribute the positive amount $d_j 10^{-j}$ to a sum of nonnegative terms. Hence zero has just the expansion $0.\overline{0}$.

Conclusion. The two claimed alternative expansions follow from the geometric tail identity. Every positive terminating decimal has the indicated second expansion; zero is the exceptional case.

16.3 Exercise 16.3: Strict comparison of convergent nonnegative series

What we must show. Suppose $\sum a_n$ and $\sum b_n$ converge, all their terms are nonnegative, $a_n \le b_n$ for every n, and $a_m < b_m$ at at least one index m. Prove the inequality of their sums is strict.

Plan. Keep the positive gap at that one index when passing from finite sums to infinite sums. A strict inequality at each partial sum alone would not guarantee a strict inequality of limits.

Name the sums and the fixed positive gap.

Write

$$A = \sum_{n=1}^{\infty} a_n, \qquad B = \sum_{n=1}^{\infty} b_n, \qquad \delta = b_m - a_m > 0.$$

Both A and B are finite by hypothesis. For every $N \ge m$,

$$\sum_{n=1}^{N} b_n - \sum_{n=1}^{N} a_n = \sum_{n=1}^{N} (b_n - a_n) \ge b_m - a_m = \delta.$$

The inequality holds because all the other differences are nonnegative; deleting them cannot increase the sum.

Take the limit with the gap still present.

The difference rule for limits gives

$$B - A = \lim_{N\to\infty} \left(\sum_{n=1}^{N} b_n - \sum_{n=1}^{N} a_n \right) \ge \delta > 0.$$

Here the weak inequality is preserved by passage to finite limits, and the final strict inequality comes from the fixed number δ, which does not shrink with N.

Conclusion. $\displaystyle\sum_{n=1}^{\infty} a_n < \sum_{n=1}^{\infty} b_n.$

16.4 Exercise 16.4: Converting six repeating decimals to fractions

What we must show. Express each given repeating decimal as a fraction of integers.

Plan. Group the repeated block as an integer and use a geometric series. A block of r digits with integer value D, repeated immediately after the decimal point, represents

$$\frac{D}{10^r} + \frac{D}{10^{2r}} + \cdots = \frac{D10^{-r}}{1 - 10^{-r}} = \frac{D}{10^r - 1}.$$

A leading zero counts as a digit of the block: the block 02 has length two and integer value 2.

(a) The decimal $0.\overline{2}$.

Here the block has length one and value 2, so

$$0.\overline{2} = \frac{2}{10 - 1} = \boxed{\frac{2}{9}}.$$

(b) The decimal $0.0\overline{2}$.

Only the digit 2 repeats; the first decimal digit is the single nonrepeating zero. Moving the decimal point one place right gives $10x = 0.\overline{2} = 2/9$, where $x = 0.0\overline{2}$. Therefore

$$x = \frac{1}{10}\frac{2}{9} = \frac{2}{90} = \boxed{\frac{1}{45}}.$$

This differs from part (c), where the zero repeats as part of the block.

(c) The decimal $0.\overline{02}$.

The repeated block has length two, so

$$0.\overline{02} = \frac{2}{10^2 - 1} = \boxed{\frac{2}{99}}.$$

(d) The decimal $3.\overline{14}$.

Separate the integer part before evaluating the length-two block:

$$3.\overline{14} = 3 + \frac{14}{99} = \frac{297 + 14}{99} = \boxed{\frac{311}{99}}.$$

(e) The decimal $0.\overline{10}$.

The block is 10, of length two and integer value 10. Hence

$$0.\overline{10} = \boxed{\frac{10}{99}}.$$

(f) The decimal $0.\overline{1492}$.

This block has length four and integer value 1492, giving

$$0.\overline{1492} = \frac{1492}{10^4 - 1} = \boxed{\frac{1492}{9999}}.$$

Conclusion. The six fractions, in order, are 2/9, 1/45, 2/99, 311/99, 10/99, and 1492/9999.

16.5 Exercise 16.5: Decimal expansions of six rational numbers

What we must show. Find the decimal expansions of the six prescribed fractions. For terminating values, include both expansions supplied by Theorem 16.3.

(a) Expand $1/8$.

At each stage divide ten times the current remainder by 8:

$$10 = 8 \cdot 1 + 2, \qquad 20 = 8 \cdot 2 + 4, \qquad 40 = 8 \cdot 5 + 0.$$

The quotients give digits $1, 2, 5$. A zero remainder produces zeros forever. Reducing the last nonzero digit by one gives the other expansion, so

$$\boxed{\frac{1}{8} = 0.125\overline{0} = 0.124\overline{9}}.$$

(b) Expand $1/16$**.**

The successive divisions are

$$10 = 16 \cdot 0 + 10, \quad 100 = 16 \cdot 6 + 4,$$
$$40 = 16 \cdot 2 + 8, \qquad 80 = 16 \cdot 5 + 0.$$

The initial quotient zero matters: it is the first decimal digit. The digits are $0, 6, 2, 5$, followed by zeros. Thus

$$\boxed{\frac{1}{16} = 0.0625\overline{0} = 0.0624\overline{9}}.$$

(c) Expand $2/3$**.**

The first division is $20 = 3 \cdot 6 + 2$. The new remainder is the original remainder 2, so the same digit 6 and remainder 2 recur at every step. Therefore

$$\boxed{\frac{2}{3} = 0.\overline{6}}.$$

As a check, the geometric-series formula gives $0.\overline{6} = 6/9 = 2/3$.

(d) Expand $7/9$**.**

Here $70 = 9 \cdot 7 + 7$. The remainder 7 returns immediately, so every digit is 7:

$$\boxed{\frac{7}{9} = 0.\overline{7}}.$$

(e) Expand $6/11$**.**

The divisions $60 = 11 \cdot 5 + 5$ and $50 = 11 \cdot 4 + 6$ return to the initial remainder 6 after the two digits $5, 4$. Hence

$$\boxed{\frac{6}{11} = 0.\overline{54}}.$$

The block formula confirms $54/99 = 6/11$.

(f) Expand $22/7$**.**

First $22 = 7 \cdot 3 + 1$, giving integer part 3 and remainder 1. The next six divisions are

$$10 = 7 \cdot 1 + 3, \quad 30 = 7 \cdot 4 + 2,$$
$$20 = 7 \cdot 2 + 6, \quad 60 = 7 \cdot 8 + 4,$$
$$40 = 7 \cdot 5 + 5, \quad 50 = 7 \cdot 7 + 1.$$

The remainder has returned to 1, so the six-digit block repeats:

$$\boxed{\frac{22}{7} = 3.\overline{142857}}.$$

Account for all possible expansions.

The last four displayed expansions have neither an eventual tail of zeros nor an eventual tail of nines. Theorem 16.3 says they are unique. The first two are positive terminating decimals, so the two representations displayed in each of (a) and (b) are their only representations.

Conclusion. The expansions are the six boxed answers above, including both representations in the two terminating cases.

16.6 Exercise 16.6: The cyclic decimal blocks for the sevenths

What we must show. Find the expansions of $1/7, 2/7, \ldots, 6/7$ and explain the pattern in their repeating blocks.

Record the long division rule once for each possible nonzero remainder.

For a current remainder r, the next digit d and next remainder r' satisfy $10r = 7d + r'$ with $0 \le r' < 7$. The six possible nonzero remainders give

r	1	2	3	4	5	6
d	1	2	4	5	7	8
r'	3	6	2	5	1	4

For instance, the entry $r = 3$ comes from $30 = 7 \cdot 4 + 2$, so the next digit is 4 and the next remainder is 2.

Follow the cycle starting from remainder one.

The remainder rule traces

$$1 \longmapsto 3 \longmapsto 2 \longmapsto 6 \longmapsto 4 \longmapsto 5 \longmapsto 1.$$

The digits produced on these six steps are respectively $1, 4, 2, 8, 5, 7$. All six remainders in the cycle are distinct, so returning to the initial remainder takes exactly six steps.

Start the same cycle at each of the other remainders.

The fraction $a/7$ begins with remainder a, for $a = 1, \ldots, 6$. Starting the displayed cycle at that remainder gives

Fraction	Remainders before the successive digits	Expansion
$1/7$	$1, 3, 2, 6, 4, 5$	$0.\overline{142857}$
$2/7$	$2, 6, 4, 5, 1, 3$	$0.\overline{285714}$
$3/7$	$3, 2, 6, 4, 5, 1$	$0.\overline{428571}$
$4/7$	$4, 5, 1, 3, 2, 6$	$0.\overline{571428}$
$5/7$	$5, 1, 3, 2, 6, 4$	$0.\overline{714285}$
$6/7$	$6, 4, 5, 1, 3, 2$	$0.\overline{857142}$

Each row returns to its initial remainder after the six digits, so the same block repeats forever. These expansions are unique because none has an eventual tail of zeros or nines.

Conclusion. The six blocks are cyclic rotations of 142857. Long division uses the same cycle of six nonzero remainders, with a different starting position for each fraction.

16.7 Exercise 16.7: The decimal formed by concatenating integers

What we must show. Decide whether the number

$$c = 0.12345678910111213141516171819202122223\cdots$$

is rational. Here its digits are obtained by writing $1, 2, 3, \ldots$ in their ordinary decimal notation and concatenating those strings.

Plan. Prove this digit string is not eventually periodic. It has arbitrarily long strings of zeros arbitrarily far out, but it also has nonzero digits arbitrarily far out. Both facts are needed: a decimal ending in zeros would have the first property alone.

Find long zero strings after any prescribed position.

For every positive integer m, the integer 10^m appears in the concatenation. Its ordinary decimal notation is the digit 1 followed by m zeros. As m increases, the starting positions of these integers go arbitrarily far out: before writing 10^m, we have already written the $10^m - 1$ preceding positive integers, each of which contributes at least one digit. Thus, after any specified digit position and for any specified length p, there is a string of at least p consecutive zeros. Choose $m \geq p$ large enough that the occurrence of 10^m starts after that position.

Find nonzero digits after any prescribed position.

The initial digit 1 in each of those occurrences of 10^m is nonzero, and these occurrences start arbitrarily far out. Consequently the decimal does not eventually consist only of zeros. It does not eventually consist only of nines either, since zero strings also occur arbitrarily far out.

Rule out an eventual repeating block.

Suppose the digits were periodic with period $p \geq 1$ after some position N. If the repeating block contained only zeros, all digits beyond N would be zero, contradicting the preceding step. Otherwise the block contains a nonzero digit. Every run of p consecutive positions beyond N passes through every position of the repeating block once, so it must contain a nonzero digit. But the first step supplies a run of p zeros entirely beyond N. This is a contradiction in both cases.

Apply the decimal characterization of rational numbers.

The given decimal is not eventually periodic. By Theorem 16.5, a rational number has a repeating decimal expansion; by Theorem 16.3 its only possible second expansion would have tails of zeros and nines. Those tails have already been excluded here. Therefore the displayed decimal cannot represent a rational number.

Conclusion. The concatenation decimal is irrational.

16.8 Exercise 16.8: Constructing a number outside a proposed list

What we must show. Given any sequence (s_n) in $(0,1)$, with selected expansions $s_n = 0.d_1^{(n)}d_2^{(n)}\cdots$, define

$$e_n = \begin{cases} 6, & d_n^{(n)} \neq 6, \\ 7, & d_n^{(n)} = 6. \end{cases}$$

Show that $0.e_1e_2\cdots$ represents a number $y \in (0,1)$ different from every s_n.

Check that the new digit string represents a number.

Every e_n is either 6 or 7, hence an allowed decimal digit. Define the partial sums $y_N = \sum_{n=1}^{N} e_n 10^{-n}$. They increase and are bounded above by the convergent geometric series $\sum_{n=1}^{\infty} 7 \cdot 10^{-n} = 7/9$. The monotone sequence theorem gives a finite limit y, which by the definition of a decimal is the number $0.e_1e_2\cdots$. Termwise comparison of the finite partial sums and passage to limits give

$$\frac{2}{3} = \sum_{n=1}^{\infty} 6 \cdot 10^{-n} \leq y \leq \sum_{n=1}^{\infty} 7 \cdot 10^{-n} = \frac{7}{9}.$$

Because $0 < 2/3 \leq 7/9 < 1$, these bounds imply $y \in (0,1)$.

Verify the diagonal difference at every index.

Fix n. If $d_n^{(n)} \neq 6$, the construction gives $e_n = 6$, so $e_n \neq d_n^{(n)}$. If $d_n^{(n)} = 6$, it gives $e_n = 7$, so again $e_n \neq d_n^{(n)}$. Thus the new expansion differs from the selected expansion of s_n at least at its nth decimal digit.

Explain why different strings cannot secretly represent the same number.

Different decimal strings can represent the same real number in the zero-tail/nine-tail case of Theorem 16.3. That issue must be checked rather than ignored. Every digit of the expansion of y is 6 or 7, so this expansion ends neither in zeros nor in nines. Theorem 16.3 therefore makes it the unique expansion of y. If $y = s_n$, the selected expansion of s_n would be a second expansion of y differing at digit n, which is impossible. Hence $y \neq s_n$ for every n.

Interpret the conclusion for a supposed enumeration.

If a sequence listed every number in $(0,1)$, the number y constructed from that sequence would belong to $(0,1)$ and yet would be absent from the list. This contradiction proves that no such listing exists.

Conclusion. The constructed y lies in $(0,1)$ and differs from every listed term. Thus $(0,1)$ is uncountable.

16.9 Exercise 16.9: Existence of Euler's constant

What we must show. For

$$H_n = \sum_{k=1}^{n} \frac{1}{k}, \qquad \gamma_n = H_n - \log n = H_n - \int_1^n \frac{dt}{t},$$

prove (a) (γ_n) decreases, (b) $0 < \gamma_n \leq 1$, and (c) its limit exists and is finite. The logarithm here is natural.

(a) Subtract two neighboring terms without discarding the logarithm.

Since $H_{n+1} = H_n + 1/(n+1)$, we have

$$\begin{aligned}\gamma_n - \gamma_{n+1} &= H_n - \log n - H_n - \frac{1}{n+1} + \log(n+1) \\ &= \int_n^{n+1} \frac{dt}{t} - \frac{1}{n+1} \\ &= \int_n^{n+1} \left(\frac{1}{t} - \frac{1}{n+1}\right) dt.\end{aligned}$$

For $n \le t < n+1$ the integrand is positive, because $t < n+1$ implies $1/t > 1/(n+1)$. It is continuous and positive on the interior of this interval, so its integral is strictly positive. For an explicit lower estimate, on $[n, n+1/2]$ the integrand is at least $1/(n+1/2) - 1/(n+1) > 0$, and that interval has positive length $1/2$. Hence $\gamma_n > \gamma_{n+1}$ for every n.

(b) Bound the integral between two rectangle sums.

For $t \in [k, k+1]$ with $k \ge 1$, reciprocation gives

$$\frac{1}{k+1} \le \frac{1}{t} \le \frac{1}{k}.$$

Integrating over an interval of length one preserves these bounds:

$$\frac{1}{k+1} \le \int_k^{k+1} \frac{dt}{t} \le \frac{1}{k}.$$

For $n \ge 2$, sum from $k = 1$ to $k = n-1$ to obtain

$$H_n - 1 \le \int_1^n \frac{dt}{t} \le H_n - \frac{1}{n}.$$

Subtract the middle expression from H_n. Reversal of the inequalities under subtraction gives

$$\frac{1}{n} \le \gamma_n \le 1.$$

The same conclusion holds for $n = 1$ directly: $\gamma_1 = 1 - \log 1 = 1$. In particular $\gamma_n > 0$ at every index.

(c) Apply monotone convergence with a real lower bound.

Part (a) makes (γ_n) decreasing, and part (b) bounds it below by zero. The monotone sequence theorem therefore gives a finite real limit

$$\gamma = \lim_{n\to\infty} \gamma_n \in [0, 1].$$

The fact that the two pieces H_n and $\log n$ both grow without bound does not obstruct convergence of their difference. The monotonicity and bounds control that difference directly.

Conclusion. The sequence is strictly decreasing, satisfies $0 < \gamma_n \le 1$, and has a finite limit called Euler's constant.

16.10 Exercise 16.10: Irrationality of the indicated powers and roots of pi

What we must show. Starting with the irrationality of π^2 established in Example 6(b), determine what follows for π, $\sqrt{\pi}$, and $\sqrt[3]{\pi}$. Also address π^4 and distinguish the additional fact needed for that last conclusion.

Recall the elementary closure property of rational powers.

If $q = a/b$ is rational, with $a, b \in \mathbb{Z}$ and $b \neq 0$, then for every positive integer m,

$$q^m = \frac{a^m}{b^m} \in \mathbb{Q},$$

since the numerator and denominator are integers and $b^m \neq 0$. Consequently, if a positive integer power of a number is irrational, the number itself must be irrational.

Apply this property to π and to its square root.

If π were rational, its square would be rational, contrary to the given result. Therefore π is irrational. If $\sqrt{\pi}$ were rational, its square would be π and would be rational, giving the same contradiction. Equivalently, its fourth power is π^2, which is already known to be irrational. Thus $\sqrt{\pi}$ is irrational too.

Apply the same argument to the cube root.

If $\sqrt[3]{\pi}$ were rational, its cube π would be rational. Since π is irrational, this is impossible. One can also apply the closure property directly to the sixth power: $(\sqrt[3]{\pi})^6 = \pi^2$. Hence $\sqrt[3]{\pi}$ is irrational.

Explain what the premise alone says about π^4.

The implication just used runs from irrationality of a power to irrationality of its base. It cannot be reversed: the square of an irrational number can be rational. For a concrete comparison, let $x = \sqrt[4]{2}$. Then $x^2 = \sqrt{2}$ is irrational, but $x^4 = 2$ is rational. Thus the statement "x^2 is irrational" does not by itself imply "x^4 is irrational." In particular, the given irrationality of π^2 alone does not settle the question about π^4 by this rational-power argument.

Determine the actual status of π^4 using the stronger fact stated in the text.

Example 6(c) additionally states the deeper theorem that π is not algebraic. This means it is not a zero of any nonzero polynomial with integer coefficients. That stated theorem does settle the question: if $\pi^4 = a/b$ were rational, where $a, b \in \mathbb{Z}$ and $b \neq 0$, then

$$b\pi^4 - a = 0.$$

The polynomial $P(X) = bX^4 - a$ has integer coefficients and is nonzero because its leading coefficient b is nonzero. The displayed equation would make π algebraic, contradicting Example 6(c). Therefore π^4 is irrational.

This final step cites the stronger theorem expressly stated in the chapter; it is not a consequence of the irrationality of π^2 alone.

Conclusion. π, $\sqrt{\pi}$, and $\sqrt[3]{\pi}$ are irrational by elementary rational-power arguments from π^2. The number π^4 is also irrational, using the stronger nonalgebraicity result stated in Example 6(c).